

# GAMMA-RAY BURST CLASSIFICATION

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# $T_{90}$ DISTRIBUTION

- **Gamma-ray bursts** (GRBs) are usually classified into two different categories, according to their duration: **short GRBs** and **long GRBs**. The distribution of the burst duration,  $T_{90}$ , is usually modelled as the **weighted sum of two log-normal distributions**:

$$p(T_{90}) = w_1 \mathcal{N}(\log T_{90} | \mu_1, \sigma_1^2) + (1 - w_1) \mathcal{N}(\log T_{90} | \mu_2, \sigma_2^2)$$

- The available data are  $x = \log T_{90}$ ; the distribution becomes a **weighted sum of two normal distributions**:

$$p(x) = w_1 \mathcal{N}(x | \mu_1, \sigma_1^2) + (1 - w_1) \mathcal{N}(x | \mu_2, \sigma_2^2)$$

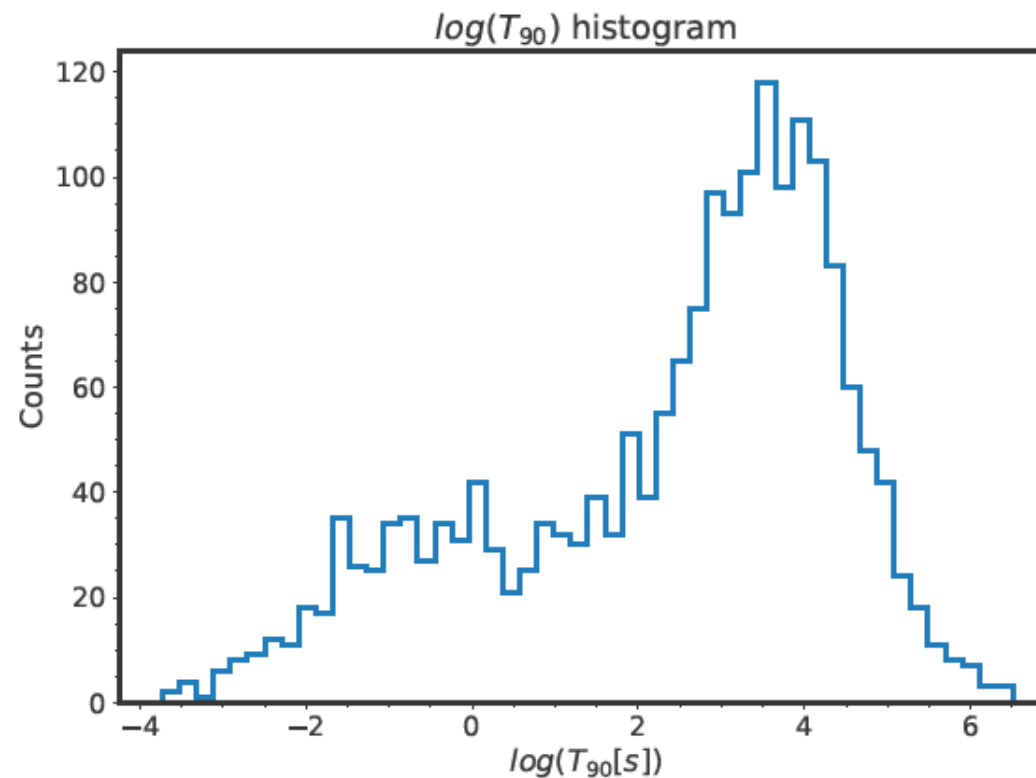
$$p(T_{90}) = p(\log T_{90}) \frac{d(\log T_{90})}{dT_{90}} = p(\log T_{90}) / T_{90} \quad \Rightarrow \quad p(x) = p(\log T_{90}) = T_{90} p(T_{90})$$

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# PROBLEM 1A – PARAMETER ESTIMATION

- **Problem 1a:** estimate the distribution parameters while ignoring measurement uncertainties.



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# METROPOLIS ALGORITHM

The goal is to draw samples from a target distribution  $f(x)$ ; the algorithm is:

- choose starting state  $x_0$
- from current  $x_0$  draw  $x'$  from a symmetric proposal (for example Gaussian)
- compute the acceptance  $a = \min\left(1, \frac{f(x')}{f(x_0)}\right)$
- with probability  $a$  set  $x_1 = x'$ , otherwise  $x_1 = x_0$
- repeat points 2-4 to build a **Markov chain** whose **stationary distribution** is  $f(x)$



Sequence of random states where each new state depends only on the current one

Probability distribution that remains unchanged as the chain evolves (after burnin)

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# BAYES THEOREM

- $\mathbf{x} = \{\log T_{90_i}\}_{i=1}^N$
- $\boldsymbol{\theta} = (w_1, \mu_1, \delta_{12}, \sigma_1, \sigma_2)$        $\delta_{12} \stackrel{\text{def}}{=} \mu_2 - \mu_1$
- $I$  = background information (the term will be omitted in later equations for simplicity)

$$p(\boldsymbol{\theta}|\mathbf{x}, I) = \frac{p(\mathbf{x}|\boldsymbol{\theta}, I)p(\boldsymbol{\theta}|I)}{p(\mathbf{x}|I)} \propto p(\boldsymbol{\theta}|I) \prod_{i=1}^N p(x_i|\boldsymbol{\theta}, I)$$

- Data  $x_i$  are assumed independent, and each  $p(x_i|\boldsymbol{\theta}, I)$  is a mixture of two Gaussian components
  - $p(\boldsymbol{\theta}|I) = p(w_1)p(\mu_1)p(\delta_{12})p(\sigma_1)p(\sigma_2)$
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# PRIOR

- $p(w_1) = \text{Beta}(2,2)$        $\text{Beta}(\alpha, \beta) \propto x^{\alpha-1}(1-x)^{\beta-1}$

A  $\text{Beta}(2,2)$  prior assigns zero density at the boundaries 0 and 1, discouraging extreme values for  $w_1$  when only one Gaussian is present.

- $p(\mu_1) \propto \mathcal{U}(-4,7)$
- $p(\delta_{12}) \propto \mathcal{U}(0,10)$

$\delta_{12}$  prevents label switching by using the ordering constraint :  $\mu_2 \geq \mu_1$

- $p(\log \sigma_i) \propto \mathcal{U}(\log(0.1), \log(6)) \Rightarrow p(\sigma_i) \propto \frac{1}{\sigma_i}$

A uniform prior on  $\log \sigma$  ensures **scale invariance**:

$$\sigma' = a\sigma \Rightarrow p(\sigma')d\sigma' = \frac{1}{\sigma'}d\sigma' = \frac{1}{a\sigma}ad\sigma = \frac{1}{\sigma}d\sigma = p(\sigma)d\sigma$$

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# ADAPTIVE PROPOSAL

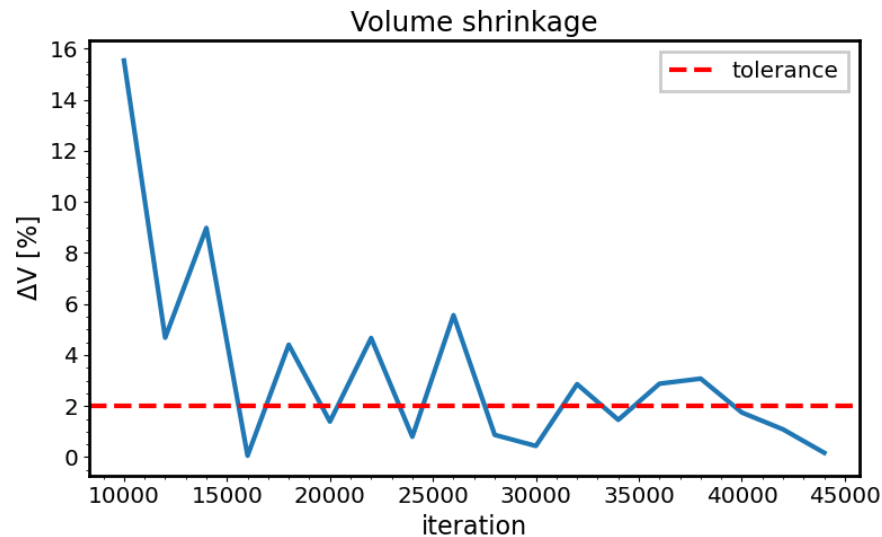
- The proposal is a multivariate Gaussian.
- If the covariance is  $I_{5 \times 5}$  or  $0.01 \times I_{5 \times 5}$ , the autocorrelation decays to 0 after lag 1000.

$$\rho(\tau) = \frac{\mathbb{E}[(x(t) - \bar{x})(x(t + \tau) - \bar{x})]}{\mathbb{E}[(x(t) - \bar{x})^2]}$$

- Two runs:
    1. **Warm-up**: adapt the proposal (samples not valid, chain is non-Markovian).
    2. **Frozen**: draw samples using the final warm-up proposal.
  - During the warm-up run it is not possible to collect valid samples because the proposal is updated based on all previous iterations, so **the chain is not Markovian**.
  - Covariance update:  $\Sigma = sC + \varepsilon I$ , with fixed scale  $s = 1$  and small  $\varepsilon$  to prevent divergence.
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# ADAPTIVE COVARIANCE TUNING (WARM-UP)



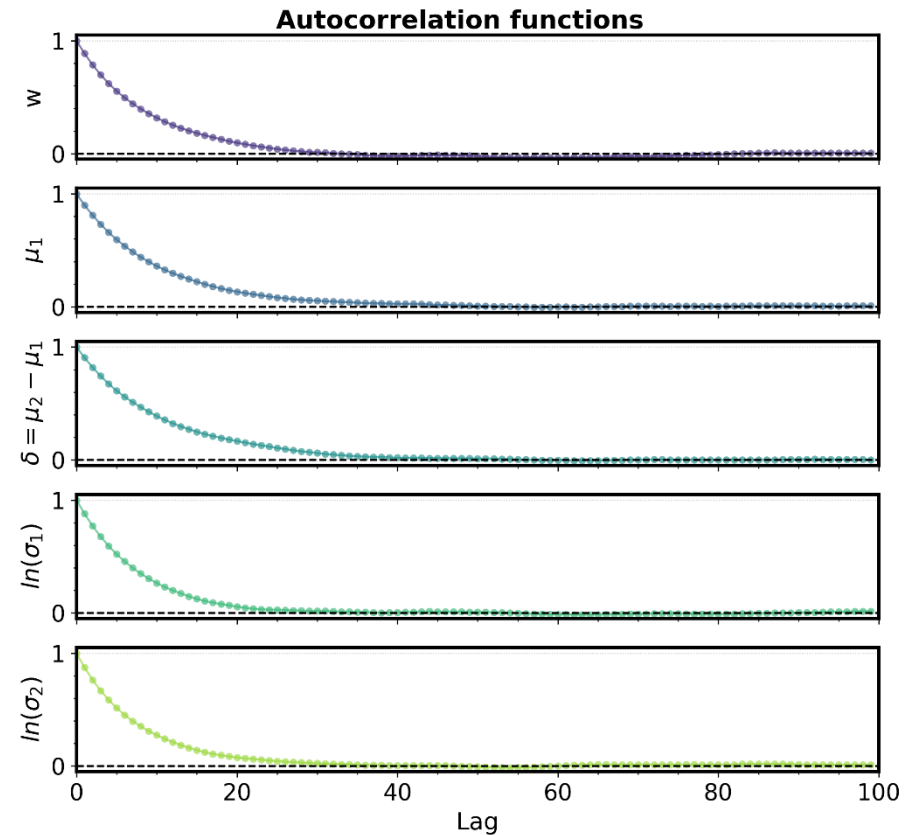
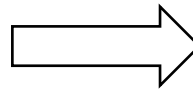
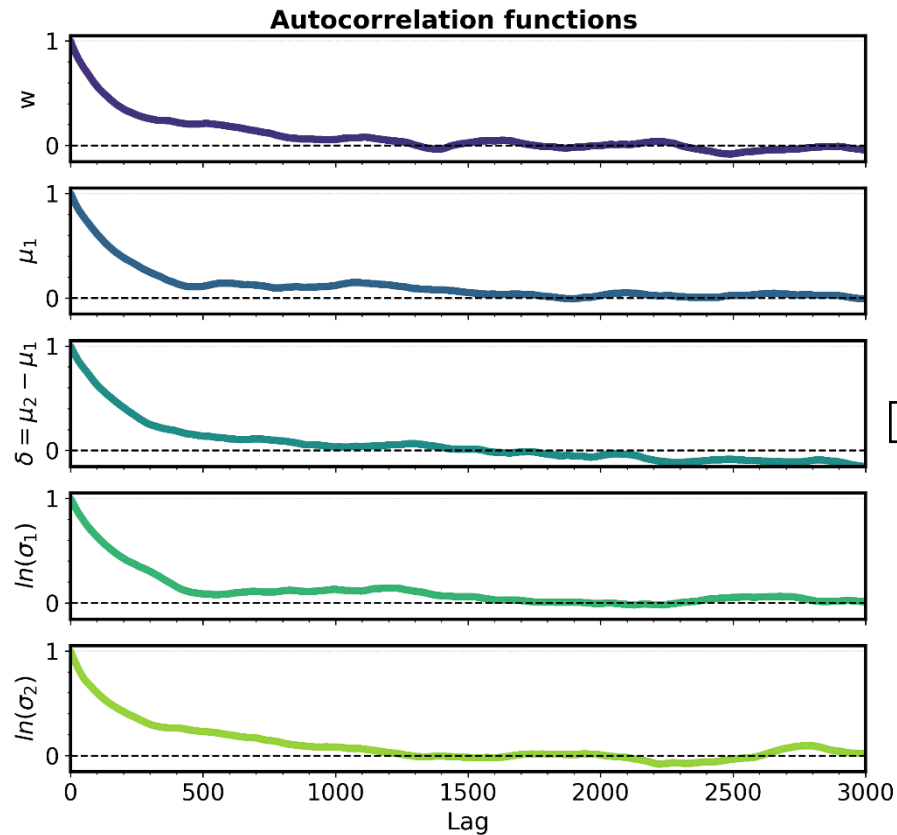
- With  $\Sigma = 0.01 \times I_{5 \times 5}$ , start a Metropolis algorithm.
- Every *adapt\_interval* steps (2000), compute:

$$\Sigma_{new} = Cov(samples_{burnin:}) + \varepsilon I$$

and set  $\Sigma = \Sigma_{new}$ .

- **Stopping criterion:** adaptation stops early if all conditions are met:
  1. Minimum number of iterations (*min\_adapt*=10000) has passed.
  2. Log-determinant change is small:  
 $|\Delta \log \det(\Sigma)| < \text{tol\_logdet} = 2 \times 10^{-2}$
  3. This small change persists for several consecutive checks (*consecutive\_stable*=3) to avoid transient fluctuations.





- **Acceptance rate:**  $0.0044 \rightarrow 0.3126$
- The data in the plots are **generated** from a mixture of Gaussians.
- The whole adaptive-proposal mechanism **is not needed**: using the sample covariance from the first (non-adaptive) run yields results that are practically identical.

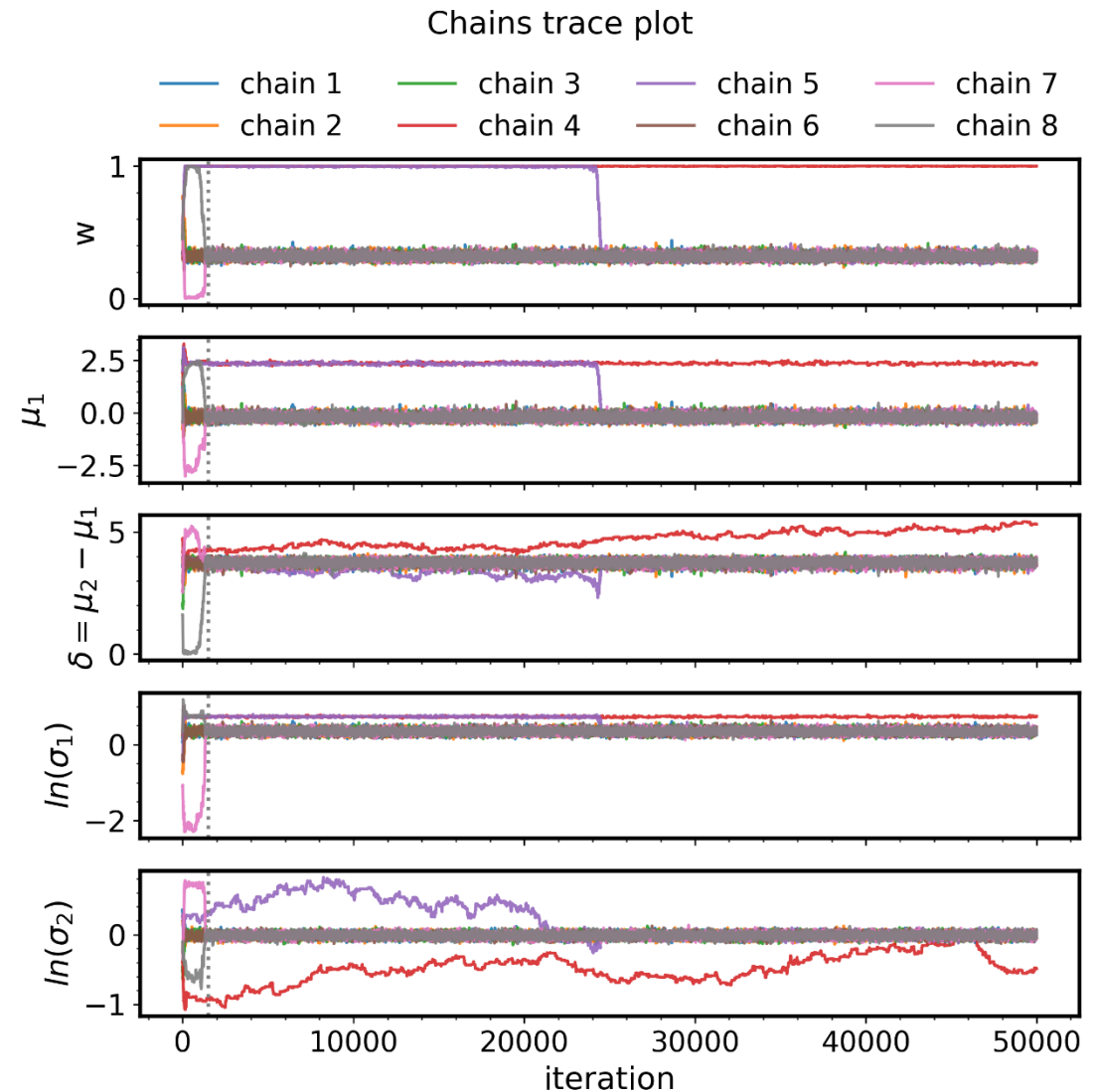
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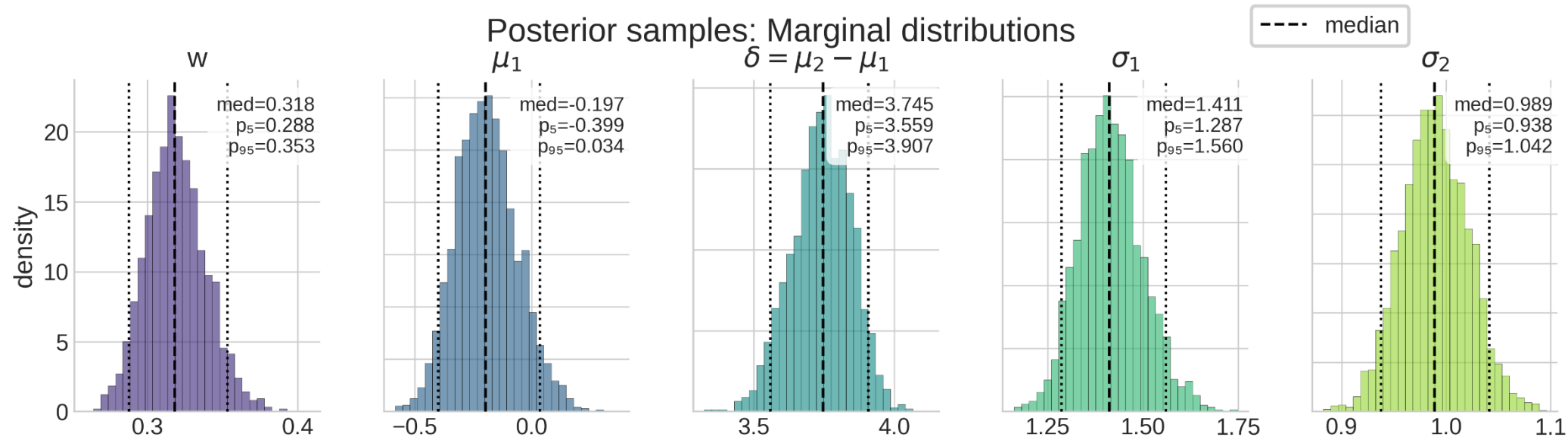
# INITIAL STATE

- The chain can get stuck near  $w = 0$  or  $w = 1$ .
  - Chains get stuck at **boundary peaks** because local proposals land in much lower posterior density and are therefore almost always rejected.
  - Peaks at  $w \approx 0$  or  $w \approx 1$  arise because there exist parameter configurations in which one mixture component fits the data by itself, making the other component effectively irrelevant.
  - To minimize the risk, the chain was initialized at  $w = 0.5$
  - Other parameters were initialized to:
    - $\sigma_i = 1$
    - $\mu_1 = p_{25}$  ( $p_i = i$ -th percentile)
    - $\delta_{12} = p_{75} - p_{25}$
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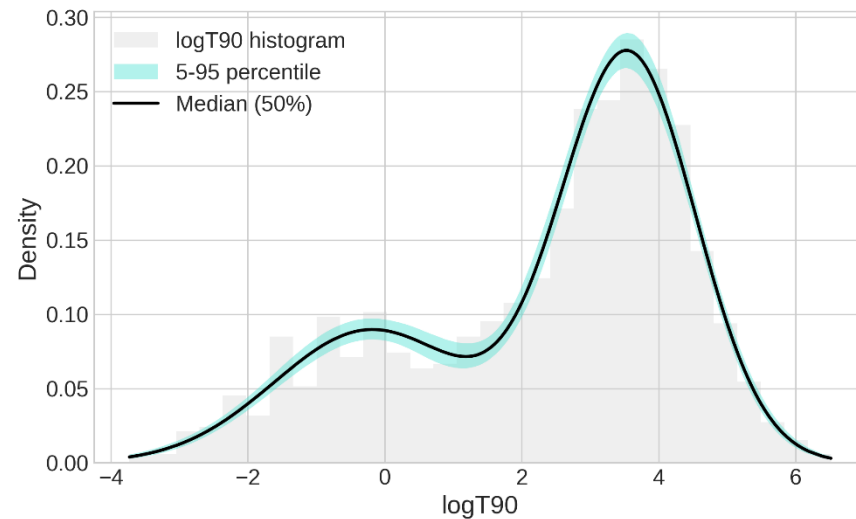
# ENSEMBLE

- Initial state validated on simulated data; real data stability uncertain.
- Ensemble of **8 chains** with **different initial states**.
- Initial states obtained by **Gaussian perturbation** (mean = base init; std = 10% of parameter range; clipped to prior boundaries).
- 2/8 chains stuck at  $w \approx 1 \rightarrow$  discarded.
- **Thinning**: first lag where autocorrelation  $< 0$  (max across parameters).
- **Burnin** = 1500.





Mean acceptance rate = 0.269  
#samples = 2975



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# PARAMETER VALUES

| Parameters            | Mean $\pm$ std    | Median $\pm[p_{16}, p_{84}]$ (68%) | $[p_5, p_{95}]$ (90%) |
|-----------------------|-------------------|------------------------------------|-----------------------|
| $w_1$                 | $0.319 \pm 0.020$ | $0.318^{+0.021}_{-0.019}$          | [0.288, 0.353]        |
| $\mu_1$ [log s]       | $-0.19 \pm 0.13$  | $-0.20^{+0.14}_{-0.12}$            | [-0.40, 0.03]         |
| $\delta_{12}$ [log s] | $3.74 \pm 0.11$   | $3.75^{+0.10}_{-0.11}$             | [3.56, 3.91]          |
| $\mu_2$ [log s]       | $3.548 \pm 0.045$ | $3.548^{+0.044}_{-0.044}$          | [3.472, 3.621]        |
| $\sigma_1$ [log s]    | $1.415 \pm 0.084$ | $1.411^{+0.087}_{-0.079}$          | [1.287, 1.560]        |
| $\sigma_2$ [log s]    | $0.989 \pm 0.032$ | $0.989^{+0.032}_{-0.031}$          | [0.938, 1.042]        |

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# PROBLEM 1B - GAUSSIAN UNCERTAINTIES

- **Problem 1b:** estimate the distribution parameters assuming **Gaussian uncertainties** on each  $\log T_{90}$ .

$$x_{obs} = x + \varepsilon \quad \varepsilon \sim \mathcal{N}(0, \sigma_{obs}^2)$$

- Marginal in  $x$ :

$$p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \int_{-\infty}^{+\infty} p(x_{obs}|x, \boldsymbol{\theta}, \sigma_{obs})p(x|\boldsymbol{\theta}, \sigma_{obs})dx$$

- The second term in the integral is independent of  $\sigma_{obs}$ . Let us assume only **one component**.

$$p(x|\boldsymbol{\theta}, \sigma_{obs}) = p(x|\boldsymbol{\theta}) = \mathcal{N}(x|\mu_1, \sigma_1^2)$$

- The first term in the integral is independent of  $\boldsymbol{\theta}$ .

$$p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \int_{-\infty}^{+\infty} \mathcal{N}(x_{obs}|x, \sigma_{obs}^2)\mathcal{N}(x|\mu_1, \sigma_1^2)dx$$

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# METHOD 1: COMPLETING THE SQUARE

- $p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \frac{1}{2\pi\sigma\sigma_{obs}} \int_{-\infty}^{+\infty} \exp\left[-\frac{1}{2}\left(\frac{(x_{obs}-x)^2}{\sigma_{obs}^2} + \frac{(x-\mu_1)^2}{\sigma_1^2}\right)\right] dx$
  - $\begin{cases} A = 1/\sigma_{obs}^2 + 1/\sigma_1^2 \\ B = x_{obs}/\sigma_{obs}^2 + \mu_1/\sigma_1^2 \\ C = x_{obs}^2/\sigma_{obs}^2 + \mu_1^2/\sigma_1^2 \end{cases} \Rightarrow \frac{(x_{obs}-x)^2}{\sigma_{obs}^2} + \frac{(x-\mu_1)^2}{\sigma_1^2} = Ax^2 - 2Bx + C = A\left(x - \frac{B}{A}\right)^2 + C - \frac{B^2}{A}$
  - $p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \frac{1}{2\pi\sigma\sigma_{obs}} \exp\left[-\frac{1}{2}\left(C - \frac{B^2}{A}\right)\right] \underbrace{\int_{-\infty}^{+\infty} \exp\left[-\frac{1}{2}A\left(x - \frac{B}{A}\right)^2\right] dx}_{= \sqrt{2\pi/A}}$
  - Replacing A, B, and C:  
$$\Rightarrow p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_{obs}^2)}} \exp\left[-\frac{1}{2} \frac{(x_{obs} - \mu_1)^2}{\sigma_1^2 + \sigma_{obs}^2}\right] = \mathcal{N}(x_{obs}|\mu_1, \sigma_1^2 + \sigma_{obs}^2)$$
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# METHOD 2: FOURIER TRANSFORM

- $p(x_{obs}|\boldsymbol{\theta}, \sigma_{obs}) = \int_{-\infty}^{+\infty} g(x_{obs} - x)f(x)dx = (f * g)(x_{obs}) = \mathcal{F}^{-1}\left(\tilde{g}(k)\tilde{f}(k)\right)$
  - $\mathcal{F}(e^{-ax^2}) = \sqrt{\frac{\pi}{a}}e^{-\frac{k^2}{4a}} \Rightarrow \begin{cases} \tilde{g}(k) = \mathcal{F}(g(x)) = \mathcal{F}\left(\frac{1}{\sqrt{2\pi\sigma_{obs}^2}}e^{-\frac{x^2}{2\sigma_{obs}^2}}\right) = e^{-\frac{1}{2}\sigma_{obs}^2k^2} \\ \tilde{f}(k) = \mathcal{F}(f(x - \mu_1)) = e^{-i\mu_1k}\mathcal{F}\left(\frac{1}{\sqrt{2\pi\sigma_1^2}}e^{-\frac{x^2}{2\sigma_1^2}}\right) = e^{-i\mu_1k}e^{-\frac{1}{2}\sigma_1^2k^2} \end{cases}$
  - $\tilde{f}(k)\tilde{g}(k) = e^{-\frac{1}{2}(\sigma_1^2 + \sigma_{obs}^2)k^2}e^{-i\mu_1k} \stackrel{\text{def}}{=} \tilde{h}(k)e^{-i\mu_1k}$
  - $\mathcal{F}^{-1}\left(\tilde{h}(k)\right) = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_{obs}^2)}}e^{-\frac{1}{2}\frac{x_{obs}^2}{\sigma_1^2 + \sigma_{obs}^2}}$   
$$\Rightarrow \mathcal{F}^{-1}\left(\tilde{f}(k)\tilde{g}(k)\right) = h(x_{obs} - \mu_1) = \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_{obs}^2)}}e^{-\frac{1}{2}\frac{(x_{obs} - \mu_1)^2}{\sigma_1^2 + \sigma_{obs}^2}} = \mathcal{N}(x_{obs}|\mu_1, \sigma_1^2 + \sigma_{obs}^2)$$
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# GAUSSIAN MIXTURE GENERALIZATION

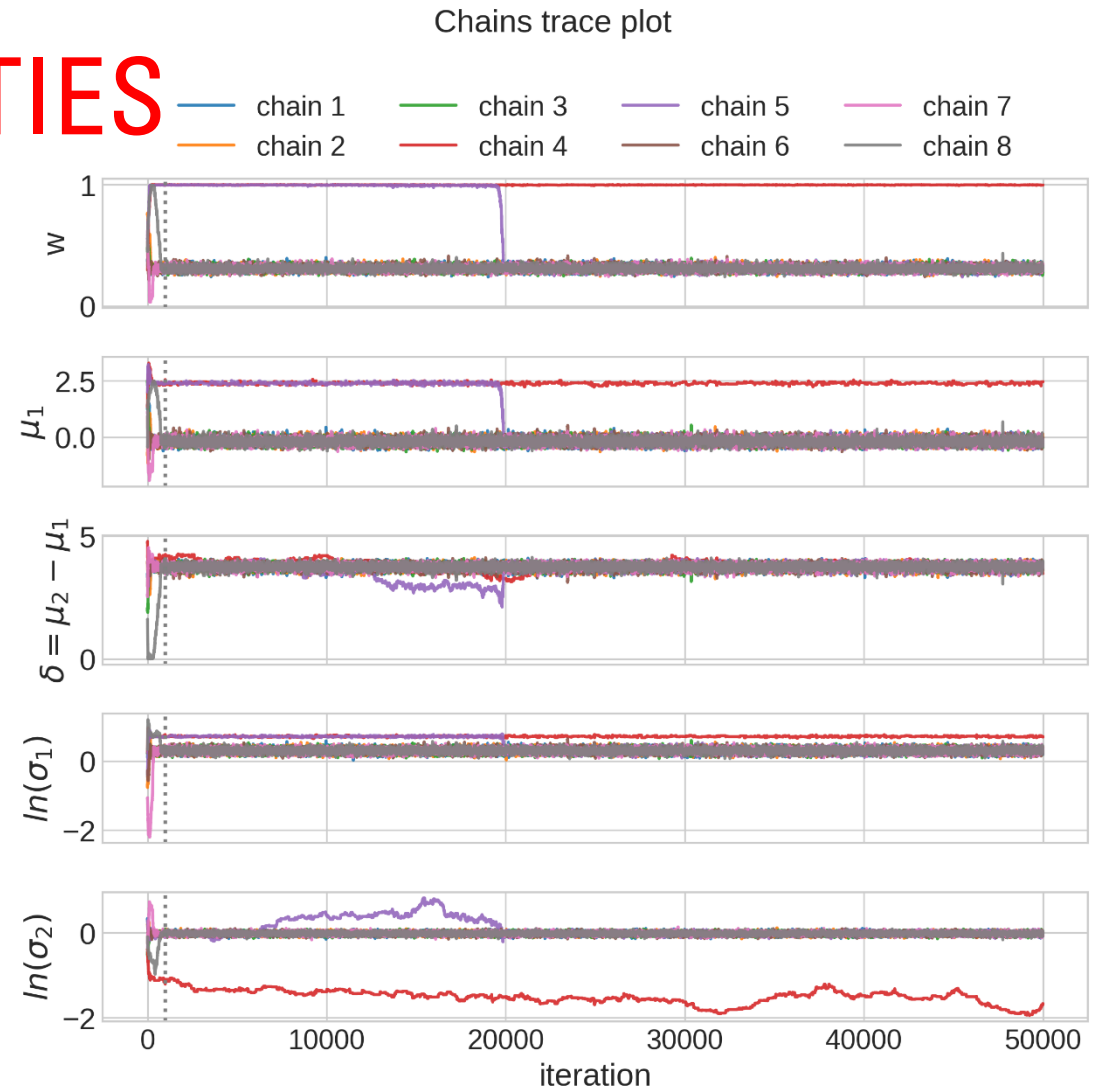
- $p(x|\boldsymbol{\theta}) = \sum_i w_i \mathcal{N}(x|\mu_i, \sigma_i^2)$

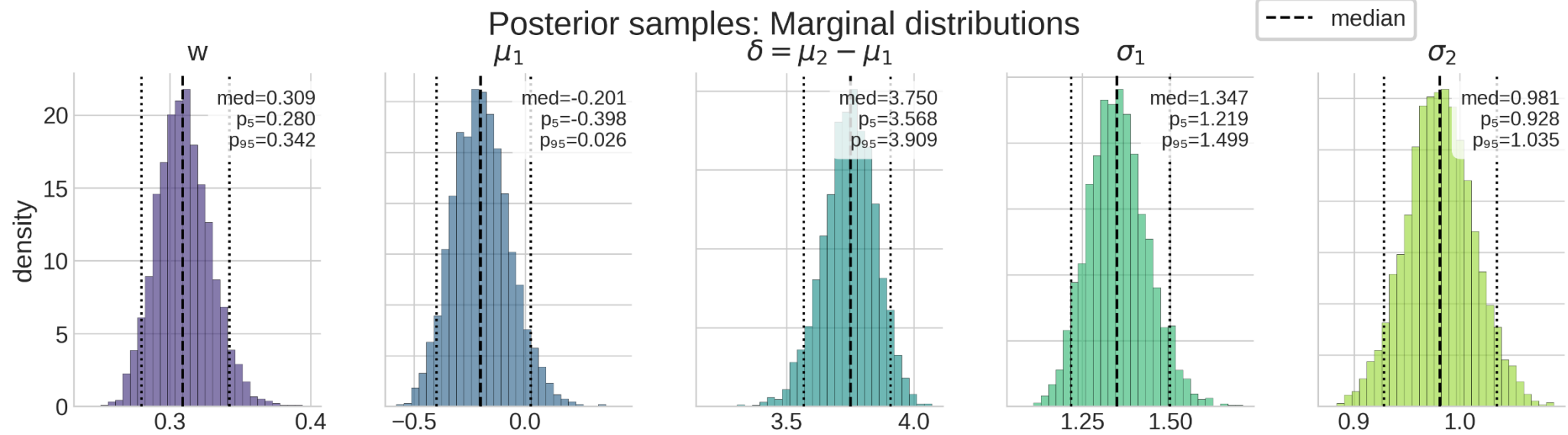
$$\begin{aligned} p(x_{obs} | \boldsymbol{\theta}, \sigma_{obs}^2) &= \int_{-\infty}^{+\infty} \mathcal{N}(x_{obs} | x, \sigma_{obs}^2) \sum_i w_i \mathcal{N}(x | \mu_i, \sigma_i^2) dx \\ &= \sum_i w_i \int_{-\infty}^{+\infty} \mathcal{N}(x_{obs} | x, \sigma_{obs}^2) \mathcal{N}(x | \mu_i, \sigma_i^2) dx \\ &= \sum_i w_i \mathcal{N}(x_{obs} | \mu_i, \sigma_i^2 + \sigma_{obs}^2) \end{aligned}$$

- The inference is analogous to Problem 1a; the values  $\sigma_{obs,i}^2$  for  $i = 1, \dots, N$  are known, so for each measurement we add  $\sigma_{obs,i}^2$  to the corresponding variance.
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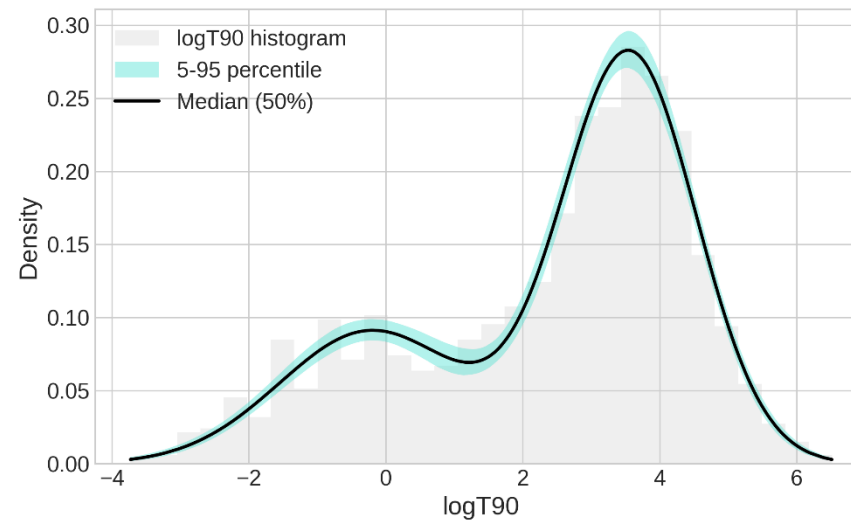
# ENSEMBLE - UNCERTAINTIES

- Initial state validated on simulated data; real data stability uncertain.
- Ensemble of **8 chains** with **different initial states**.
- Initial states obtained by **Gaussian perturbation** (mean = base init; std = 10% of parameter range; clipped to prior boundaries).
- 2/8 chains stuck at  $w \approx 1 \rightarrow$  discarded.
- **Thinning**: first lag where autocorrelation  $< 0$  (max across parameters).
- **Burnin** = 1000.





Mean acceptance rate = 0.268  
#samples = 4349



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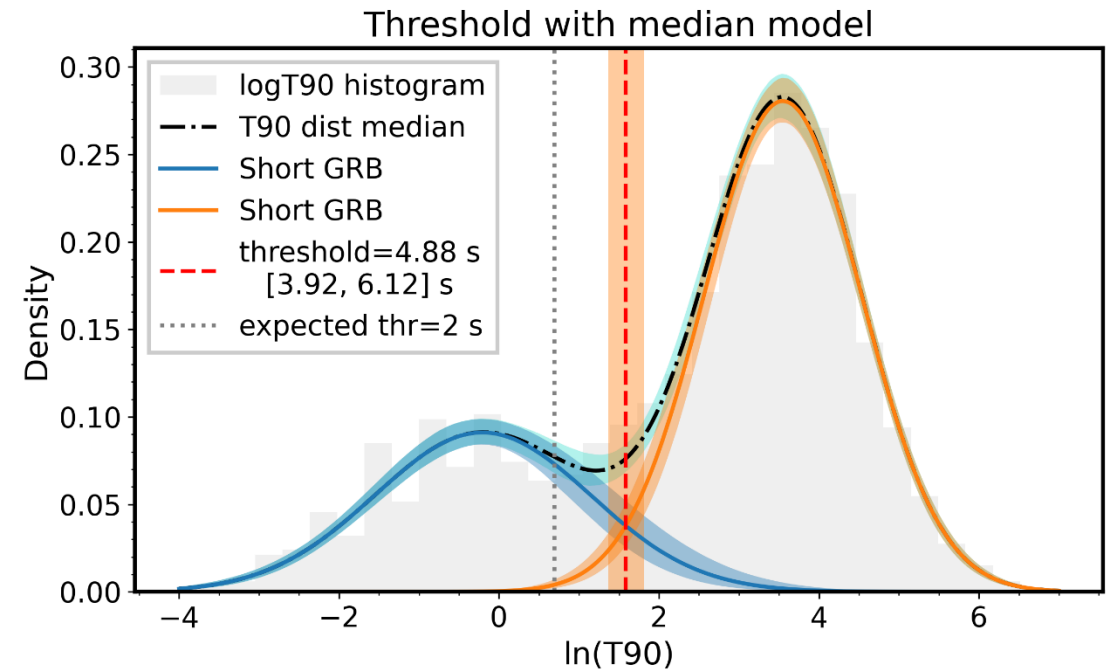
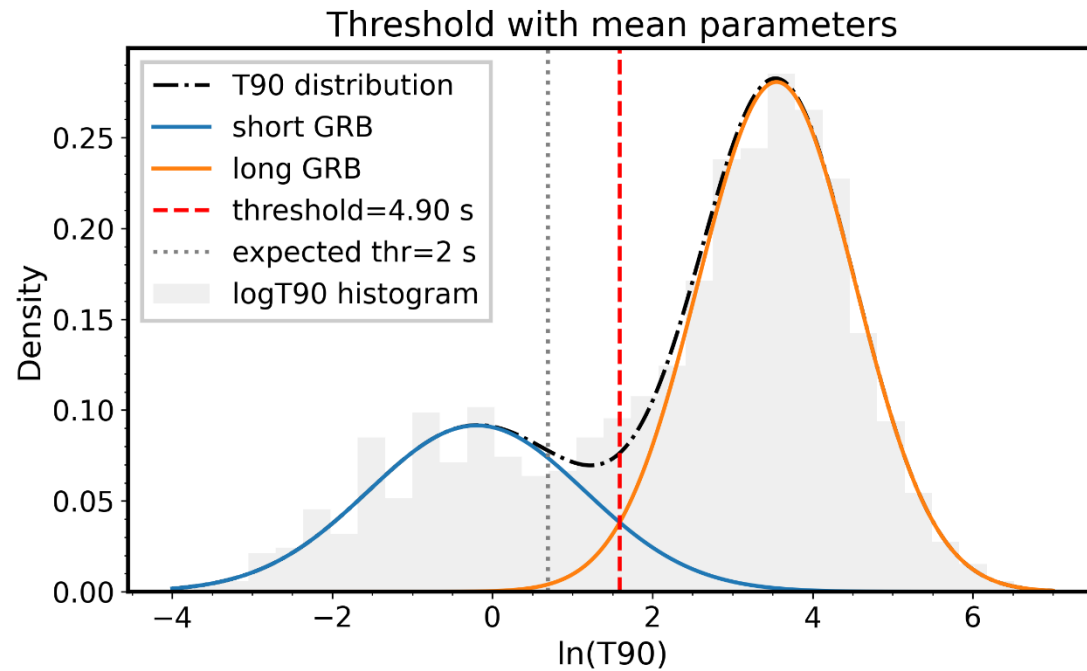
# PARAMETER VALUES - UNCERTAINTIES

| Parameters            | Mean $\pm$ std    | Median $\pm[p_{16}, p_{84}]$ (68%) | $[p_5, p_{95}]$ (90%) |
|-----------------------|-------------------|------------------------------------|-----------------------|
| $w_1$                 | $0.310 \pm 0.019$ | $0.309^{+0.019}_{-0.018}$          | [0.280, 0.342]        |
| $\mu_1$ [log s]       | $-0.20 \pm 0.13$  | $-0.20^{+0.13}_{-0.12}$            | [-0.40, 0.03]         |
| $\delta_{12}$ [log s] | $3.75 \pm 0.10$   | $3.75^{+0.10}_{-0.10}$             | [3.57, 3.91]          |
| $\mu_2$ [log s]       | $3.549 \pm 0.045$ | $3.550^{+0.044}_{-0.045}$          | [3.476, 3.623]        |
| $\sigma_1$ [log s]    | $1.351 \pm 0.085$ | $1.347^{+0.088}_{-0.080}$          | [1.219, 1.499]        |
| $\sigma_2$ [log s]    | $0.981 \pm 0.032$ | $0.981^{+0.031}_{-0.031}$          | [0.928, 1.035]        |

- Parameter estimates obtained when neglecting the uncertainty in  $x$  are **compatible** with those obtained after including that uncertainty.
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# PROBLEM 2 AND 3 - CLASSIFICATION

- **Problem 2:** compute the probability of GRB170817A ( $T = 2.0$  s) of being a short GRB or a long GRB.
- **Problem 3:** determine the threshold value for  $T_{90}$  to discriminate between short and long GRBs.

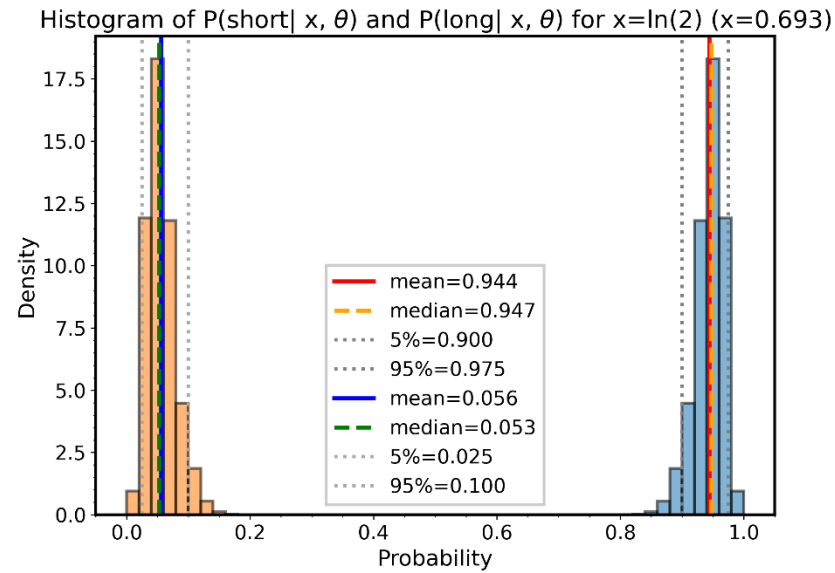
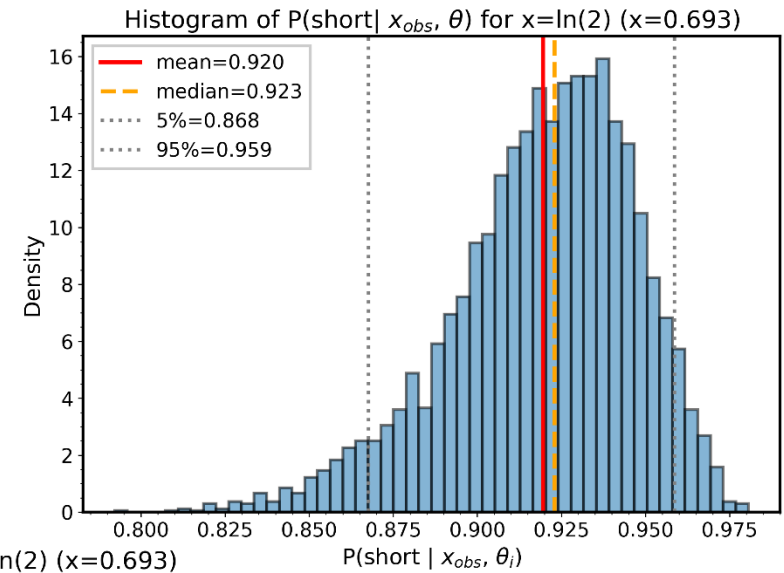
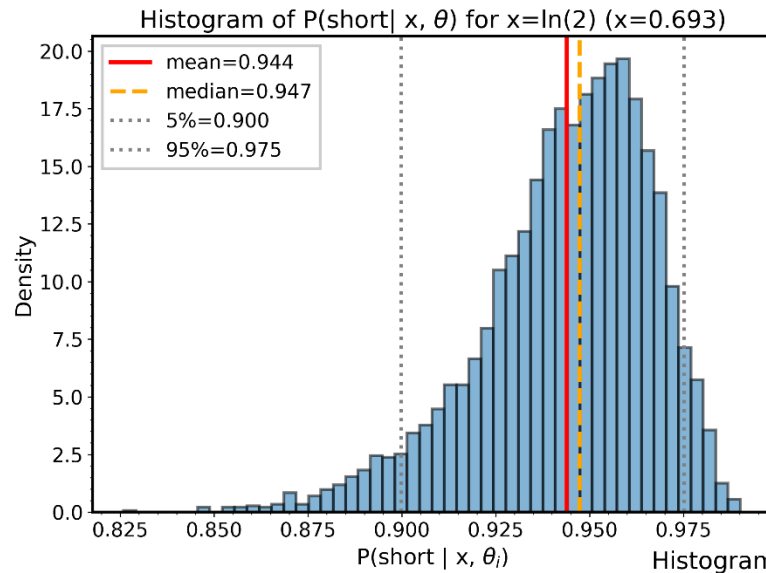


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# SHORT GRB PROBABILITY DISTRIBUTION

- $p(short|x, \boldsymbol{\theta}) = \frac{p(short|\boldsymbol{\theta})p(x|short, \boldsymbol{\theta})}{p(x|\boldsymbol{\theta})} = \frac{w_1 \mathcal{N}(x|\mu_1, \sigma_1^2)}{w_1 \mathcal{N}(x|\mu_1, \sigma_1^2) + (1-w_1) \mathcal{N}(x|\mu_2, \sigma_2^2)}$
  - $x_{obs} = x + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_{obs}^2)$   
$$\Rightarrow p(short|x_{obs}, \boldsymbol{\theta}) = \frac{w_1 \mathcal{N}(x_{obs}|\mu_1, \sigma_1^2 + \sigma_{obs}^2)}{w_1 \mathcal{N}(x_{obs}|\mu_1, \sigma_1^2 + \sigma_{obs}^2) + (1-w_1) \mathcal{N}(x_{obs}|\mu_2, \sigma_2^2 + \sigma_{obs}^2)}$$
  - The quantity is defined if the true parameters  $\boldsymbol{\theta}$  are known, but in practice we only have posterior samples  $\{\boldsymbol{\theta}_j\}_{j=1}^M$  from Metropolis.
  - Let the data be  $\boldsymbol{D} \stackrel{\text{def}}{=} \{\log T_{90_i}, \text{dlog} T_{90_i}\}_{i=1}^N$ . The value  $x_{obs}$  may or may not be included in  $\boldsymbol{D}$ .
  - $p(short|x_{obs}, \boldsymbol{D}) = \int p(short|x_{obs}, \boldsymbol{\theta})p(\boldsymbol{\theta}|\boldsymbol{D})d\boldsymbol{\theta} \approx \frac{1}{M} \sum_{j=1}^M p(short|x_{obs}, \boldsymbol{\theta}_j) \quad M = \# \text{ samples}$
  - $p(long|x_{obs}, \boldsymbol{D}) = 1 - p(short|x_{obs}, \boldsymbol{D})$
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# Distributions of GRB170817A



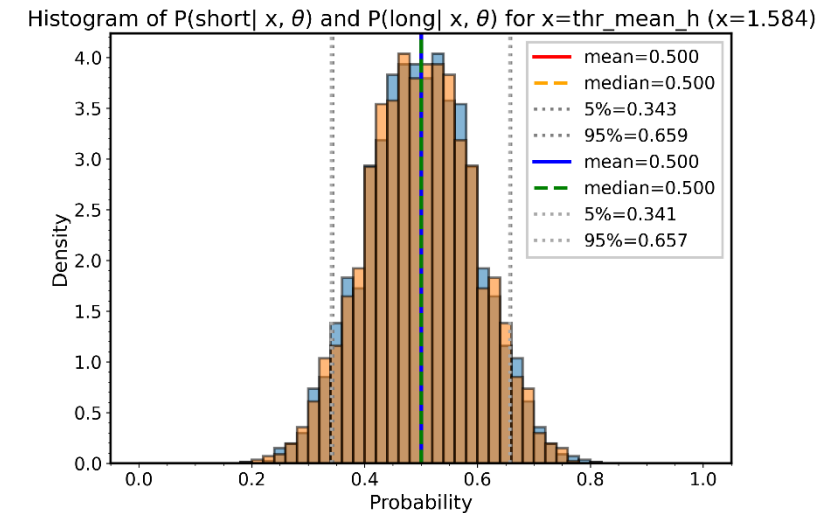
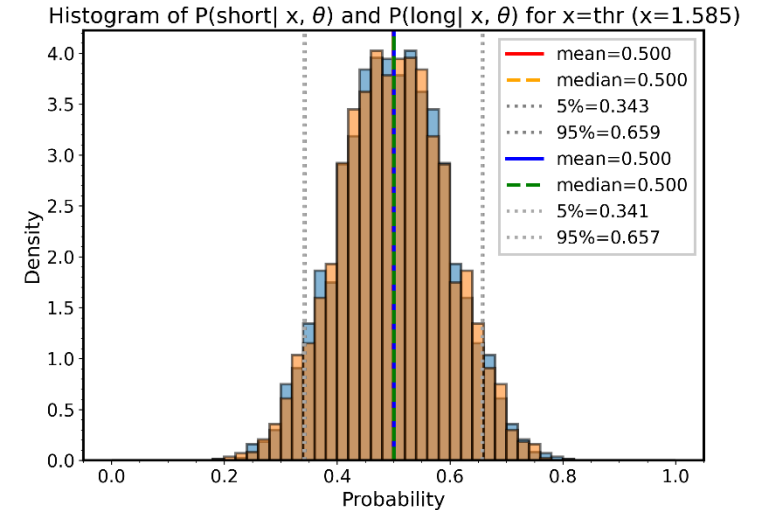


# ESTIMATE THRESHOLD

- The **graphically estimated threshold** (4.88 s) was checked and found to match expectations: the posterior mean is 0.5 and the two class distributions overlap.
- **Probability-based threshold estimation - Method 1:**
  1. Create a grid of 10,000 points over the interval  $x \in [-4, 7]$ .
  2. For each grid point  $x$  compute  $p(\text{short}|x, \theta_j)$  and extract the posterior mean  $\bar{h}(x) \approx p(\text{short}|x, \mathbf{D})$ .
  3. Extract the threshold using the **crossing method**:
    - Find two consecutive grid points  $x_1$  and  $x_2$  (with  $x_2 \geq x_1$ ) such that  $\bar{h}(x) - 0.5$  changes sign between them.
    - Estimate the threshold by **linear interpolation**:

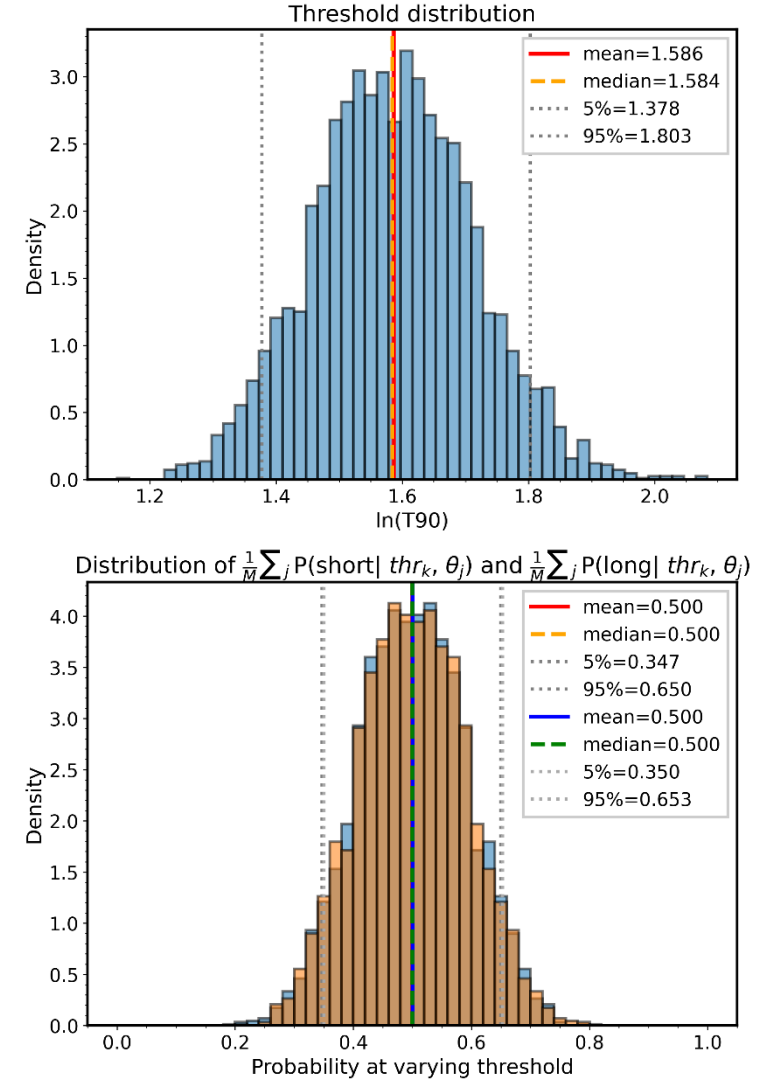
$$\text{thr\_mean\_h} = x_1 + (0.5 - \bar{h}(x_1)) \frac{x_2 - x_1}{\bar{h}(x_2) - \bar{h}(x_1)}$$

4. This yields  $\text{thr\_mean\_h} \simeq 4.87$  s



# THRESHOLD DISTRIBUTION

- **Probability-based threshold estimation - Method 2:**
  1. Create a grid of 10,000 points over the interval  $x \in [-4,7]$ .
  2. For each grid point  $x$  compute  $h_j(x) = p(\text{short}|x, \theta_j)$ .
  3. For each sample  $j$  find the threshold using the **crossing method** (the  $x$  where  $h_j(x) = 0.5$ ).
  4. This gives  $M$  thresholds  $\{thr_j\}_{j=1}^M$ ; they form the **threshold distribution**. The result is:  $thr = 4.87^{+0.57}_{-0.66} [3.96, 6.07]$  s.
- To check whether the threshold distribution matches expectations: for each  $thr_k$  compute  $p(\text{short}|thr_k, \theta_j)$  and extract the mean over the samples. Use the collection  $\{\frac{1}{M} \sum_j p(\text{short}|thr_k, \theta_j)\}_{k=1}^M$  as the **distribution of means**.



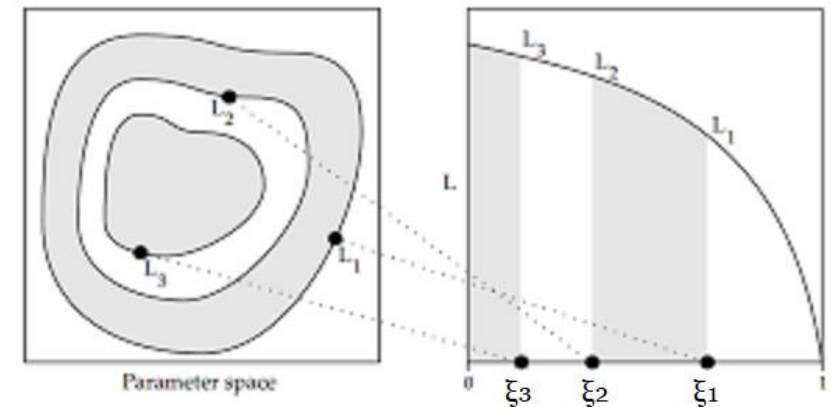
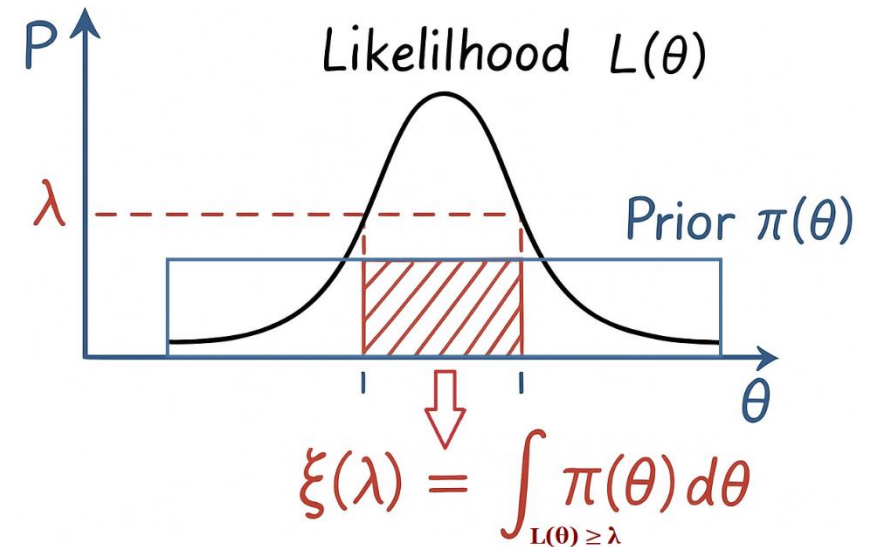
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# PROBLEM 4 – MODEL SELECTION

- **Problem 4:** Some authors propose a third class of GRBs, *intermediate*. Which of the two hypothesis is favored according to the available data?
  - $H_0$ : Mixture with **two** components  $\Rightarrow \theta_0 = (w, \mu_1, \delta, \sigma_1, \sigma_2)$
  - $H_1$ : Mixture with **three** components  $\Rightarrow \theta_1 = (v_1, v_2, \mu_1, \delta_{12}, \delta_{23}, \sigma_1, \sigma_2, \sigma_3)$
  - $\frac{p(H_1|\mathbf{D})}{p(H_0|\mathbf{D})} = \frac{p(\mathbf{D}|H_1)p(H_1)}{p(\mathbf{D}|H_0)p(H_0)}$
  - Assuming equal prior probability for the models:  $p(H_1) = p(H_0)$   
$$\Rightarrow \frac{p(H_1|\mathbf{D})}{p(H_0|\mathbf{D})} = \frac{p(\mathbf{D}|H_1)}{p(\mathbf{D}|H_0)} = \frac{\int p(\mathbf{D}|H_1, \theta_1)p(\theta_1|H_1)d\theta_1}{\int p(\mathbf{D}|H_0, \theta_0)p(\theta_0|H_0)d\theta_0} = \frac{Z_1}{Z_0} \quad \text{Bayes Factor}$$
  - **Occam's razor:** even if  $H_1$  has higher likelihood (a more complex model can fit the data better), its prior can reduce the model evidence  $Z_1$ , because  $p(\theta_1|H_1) \propto 1/V_1$  and  $V_1 \gg V_0$ , where  $V_i$  is the parameter space volume.
-

# NESTED SAMPLING (1/2)

- $Z = \int L(\theta)\pi(\theta)d\theta = \int_0^1 L(\xi)d\xi$  where  $d\xi = \pi(\theta)d\theta$
- $L(\xi)$  is **one-dimensional** and **monotonically decreasing**.
  1. Draw  $N_L$  live points from the prior and set  $\log\xi_0 = 0$ .
  2. Compute the likelihood for each live point; sort in decreasing order and pick the **lowest value**  $\log L_{min,i}$ .
  3. Discard the point with  $\log L_{min,i}$  and **draw a new sample** from the prior constrained to  $L > L_{min,i}$ .
  4. Compute  $\log\xi_i$ :
    - $\{\xi\}^i \sim \mathcal{U}(0, \xi_{i-1}) \Rightarrow \{t\}^i = \{\xi/\xi_{i-1}\}^i \sim \mathcal{U}(0,1)$
    - $t_i = \max\{t\}^i \Rightarrow t_i \sim \text{Beta}(N_L, 1) = N_L t^{N_L-1}$
    - $\mathbb{E}[\log t_i] = N_L \int_0^1 t^{N_L-1} \log t dt = -\frac{1}{N_L} \Rightarrow \log\xi_i = \log\xi_{i-1} - \frac{1}{N_L}$



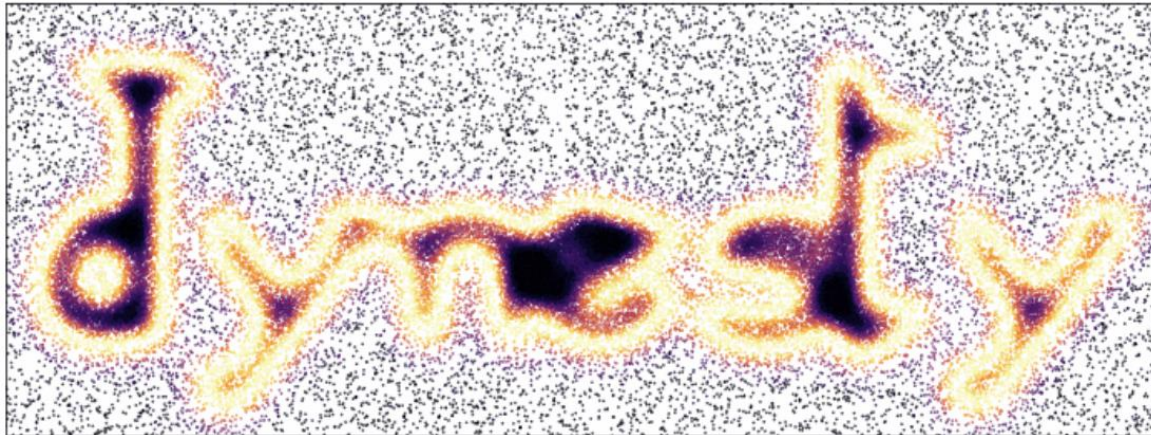
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# NESTED SAMPLING (2/2)

5. Compute **weights**:  $w_i = \xi_{i-1} - \xi_i \Rightarrow \log w_i = \log(e^{\log \xi_{i-1}} - e^{\log \xi_i})$
6. Accumulate **evidence**:  $Z_i = Z_{i-1} + w_i L_{min,i} \Rightarrow \log Z_i = \log(e^{\log Z_{i-1}} + e^{\log w_i + \log L_{min,i}})$
7. **Stopping condition**:  $\Delta \log Z_i \stackrel{\text{def}}{=} \log(Z_i + \Delta Z_i) - \log Z_i \lesssim \log(Z_i + L_{max,i} \xi_i) - \log Z_i < \varepsilon$
- $\varepsilon$  is the tolerance and  $\Delta Z_i \lesssim L_{max,i} \xi_i$  is the **estimated remaining evidence**, where  $L_{max,i}$  is the maximum value of the likelihood among the live points at iteration  $i$  and  $\xi_i$  is the remaining prior volume.
  - The condition  $\Delta Z_i \lesssim L_{max,i} \xi_i$  is an **upper bound** and is valid only if there is **not** an **extremely narrow likelihood peak** within the remaining prior volume that has not yet been discovered by the  $N_L$  live points.
8. Residual:  $Z = Z_K + \xi_K \bar{L} \Rightarrow \log Z = \log(e^{\log Z_K} + e^{\log \xi_K + \log \bar{L}})$
- $K$  indicates the final iteration and  $\bar{L}$  is the mean likelihood of the remaining live points.
9. **Posterior samples**:  $p(\theta_i) = w_i L_{min,i} / Z \Rightarrow p(\theta_i) = e^{\log w_i + \log L_{min,i} - \log Z}$
- From the available **weighted samples** ( $K + N_L = \text{\#iterations} + \text{\#residuals}$ ), draw  $M$  **unweighted posterior samples** using `numpy.random.choice` with probabilities  $p = \{p(\theta_i)\}$ .
-

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# LIBRARY: DYNESTY



- Class: NestedSampler
- #Live points  $\Rightarrow N_L = 1200$
- Tolerance  $\Rightarrow \varepsilon = 10^{-2}$
- Bound  $\Rightarrow$  Multiple ellipsoids ('multi')
- Sample  $\Rightarrow$  Random walk ('rwalk')

---

# PRIOR

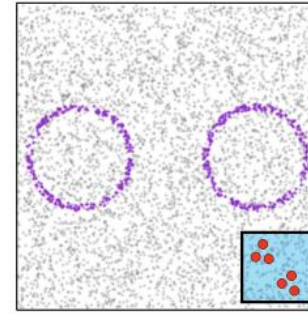
- `dynesty` uniformly draws prior samples from a multidimensional unit cube and then applies a transformation using a user-defined function.
  - For model selection, the priors must be **normalized**.
  - Weights are constructed using a **strick breaking process**:
    - $w_1 = v_1, \quad w_2 = (1 - v_1)v_2, \quad w_3 = 1 - w_1 - w_2$
    - $p(v_i) = \text{Beta}(2,2) = 6v_i(1 - v_i)$
  - $p(\mu_1) = \mathcal{U}(\min(\log T_{90}), \max(\log T_{90}))$
  - $p(\delta_{12}) = \mathcal{U}(0, \max(\log T_{90}) - \mu_1)$
  - $p(\delta_{23}) = \mathcal{U}(0, \max(\log T_{90}) - (\mu_1 + \delta_{12}))$
  - $p(\log \sigma_i) = \mathcal{U}(\log(0.1), \log(6))$
  - For  $H_0$  the priors are the same, but  $w_1 = w$  and  $w_2 = 1 - w$ .
-



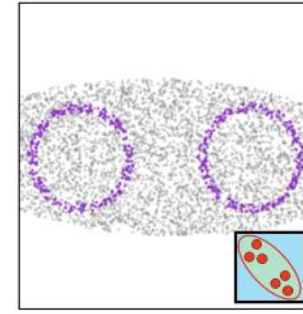
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# BOUND

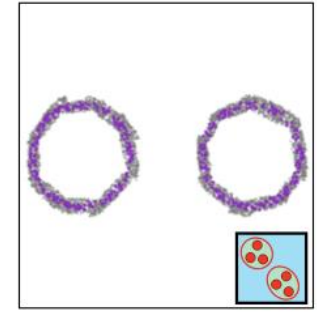
- **Bound = 'none'**
  - Inefficient: the rejection probability grows with  $L_{min,i}$ .
- **Bound = 'single'**
  - Create an ellipsoid using mean and covariance of the live points.
  - Works well for approximately Gaussian, unimodal posteriors; inefficient for multimodal posteriors.
- **Bound = 'multi'**
  - Build **multiple ellipsoids** by recursive clustering:
    1. Compute the mean and covariance of the current live points and construct an ellipsoid.
    2. Select the two points corresponding to the opposite ends of the major axis of the ellipsoid.
    3. Apply k-means using those two points ( $k=2$ ) and assign them to the two clusters.
    4. Construct an ellipsoid for each cluster.
    5. If the decrease in volume exceeds a threshold, repeat from step 2 on each ellipsoid (further subdivide); otherwise stop.



Unit Cube  
(no bound)



Single  
Ellipsoid



Multiple  
Ellipsoids

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# SAMPLE

- **Sample** = ‘unif’

- **Rejection sampling** inside the bound.

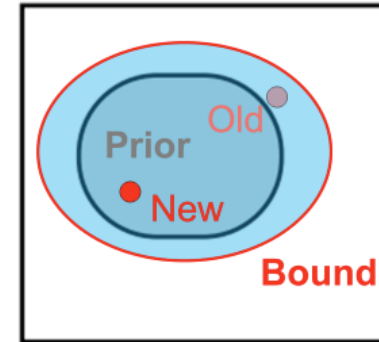
- **Sample** = ‘rwalk’

- The **proposal** is a uniform ellipsoid centered on the point to be replaced and with the same size as the bound. If *bound* = ‘multi’, randomly pick one of the ellipsoids that intersect the point.
- The **number of chain steps** per iteration is set by the parameter *walks* (default *walks* = 25).
- The **bound is updated** after  $\text{round}(0.15 \times \text{walks}) \times N_L$  likelihood calls:
  - $N_L = 1200, \text{ walks} = 25 \Rightarrow \text{round}(0.15 \times \text{walks}) \times N_L = 4500$  likelihood calls (180 iterations)
- The **proposal scale** is adapted each iteration:

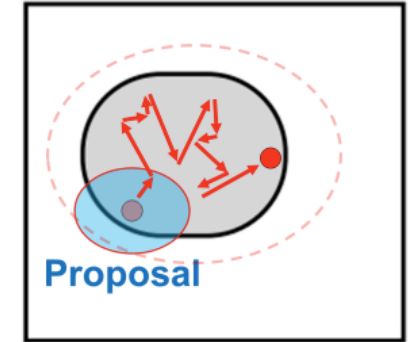
$$s_{i+1} = s_i \times \exp\left(\frac{\text{ratio} - \text{target}}{d \times \text{target}}\right) \Rightarrow \log(s_{i+1}) = \log(s_i) + \eta(\text{ratio} - \text{target}),$$

where  $\eta = 1/(d \times \text{target})$ , *ratio* is the acceptance ratio in that iteration, *target* (default is 0.5) is the desired acceptance ratio, and *d* is the parameter dimension.

Uniform

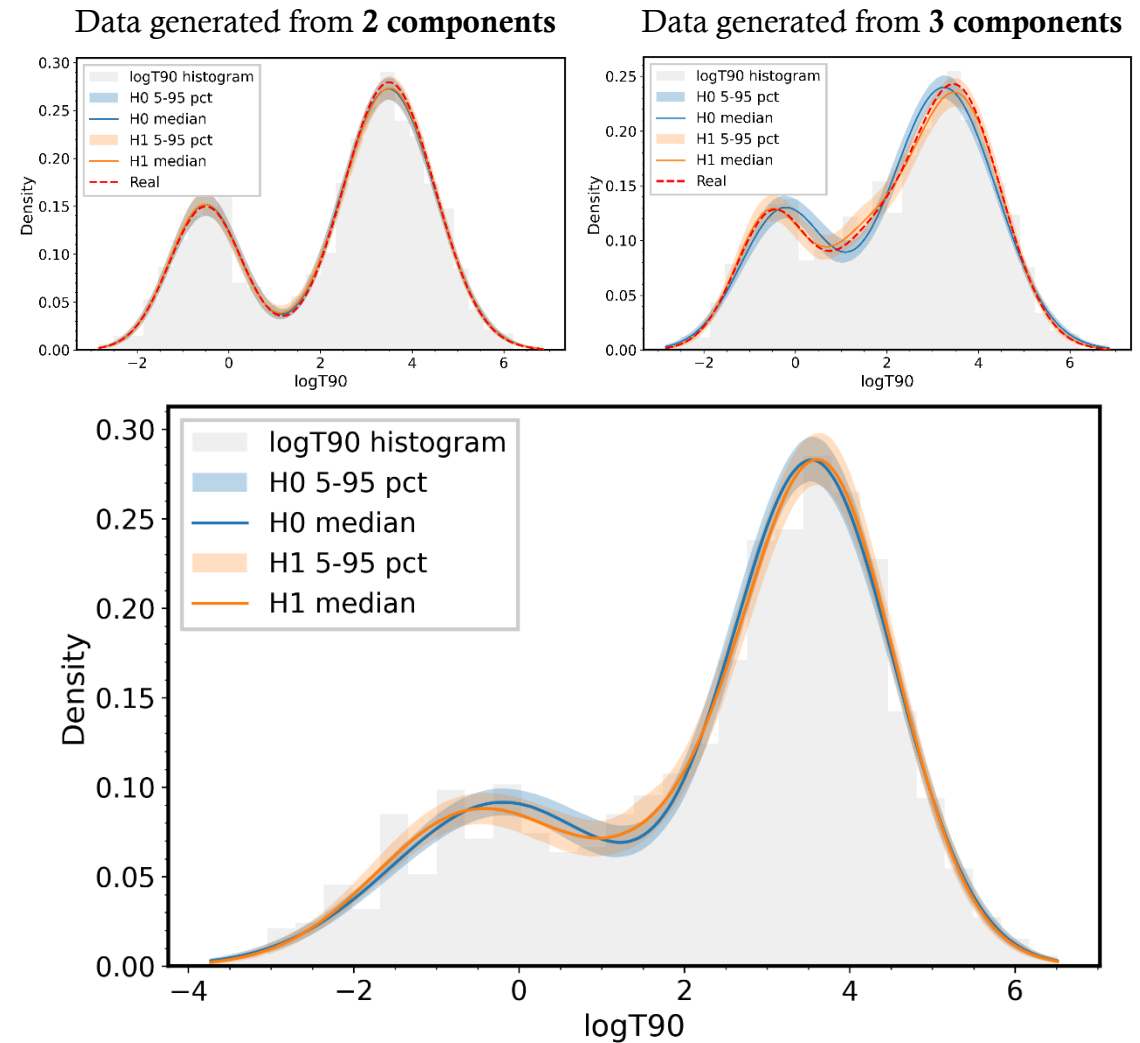


Random Walk

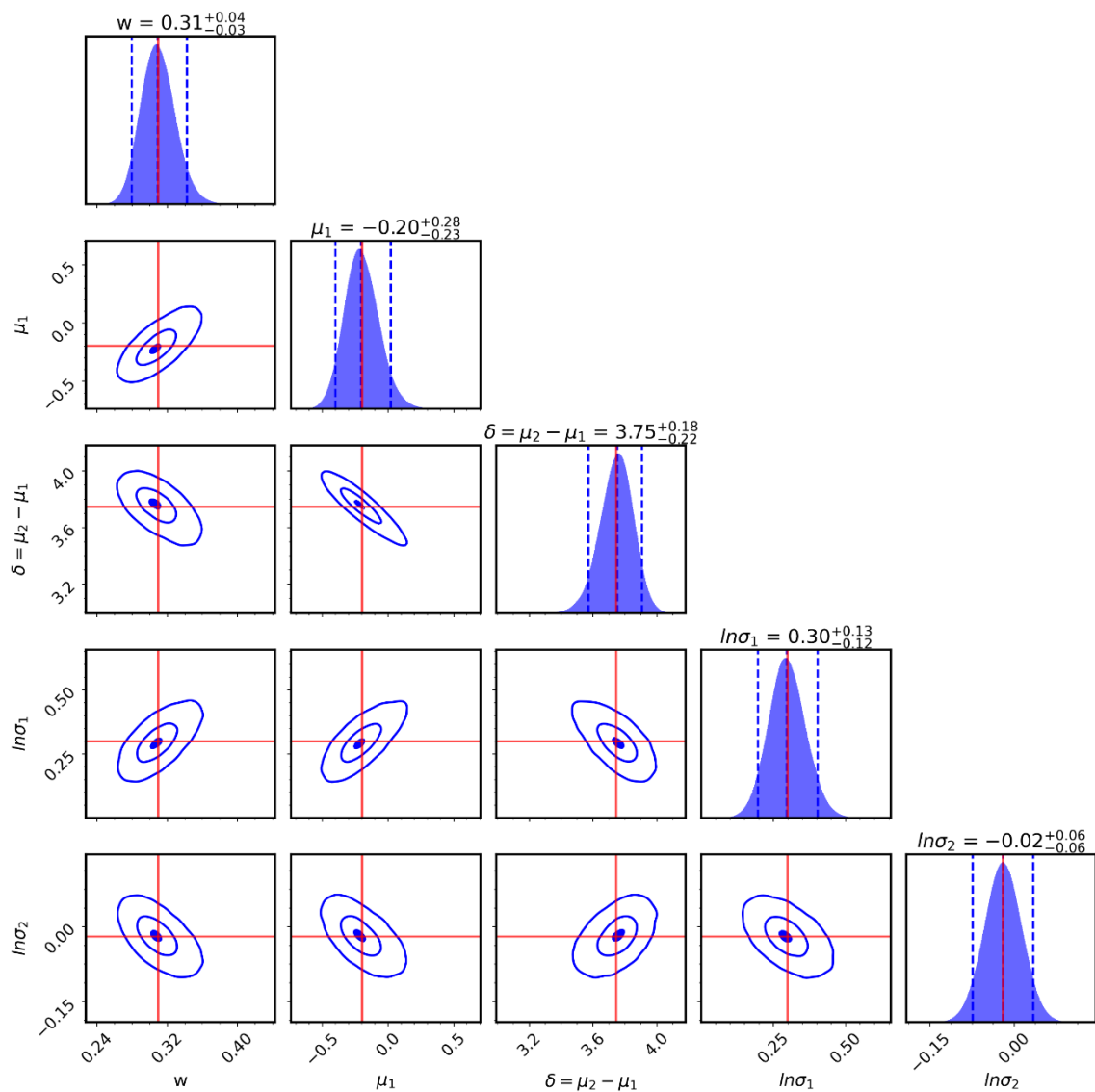


# RESULTS

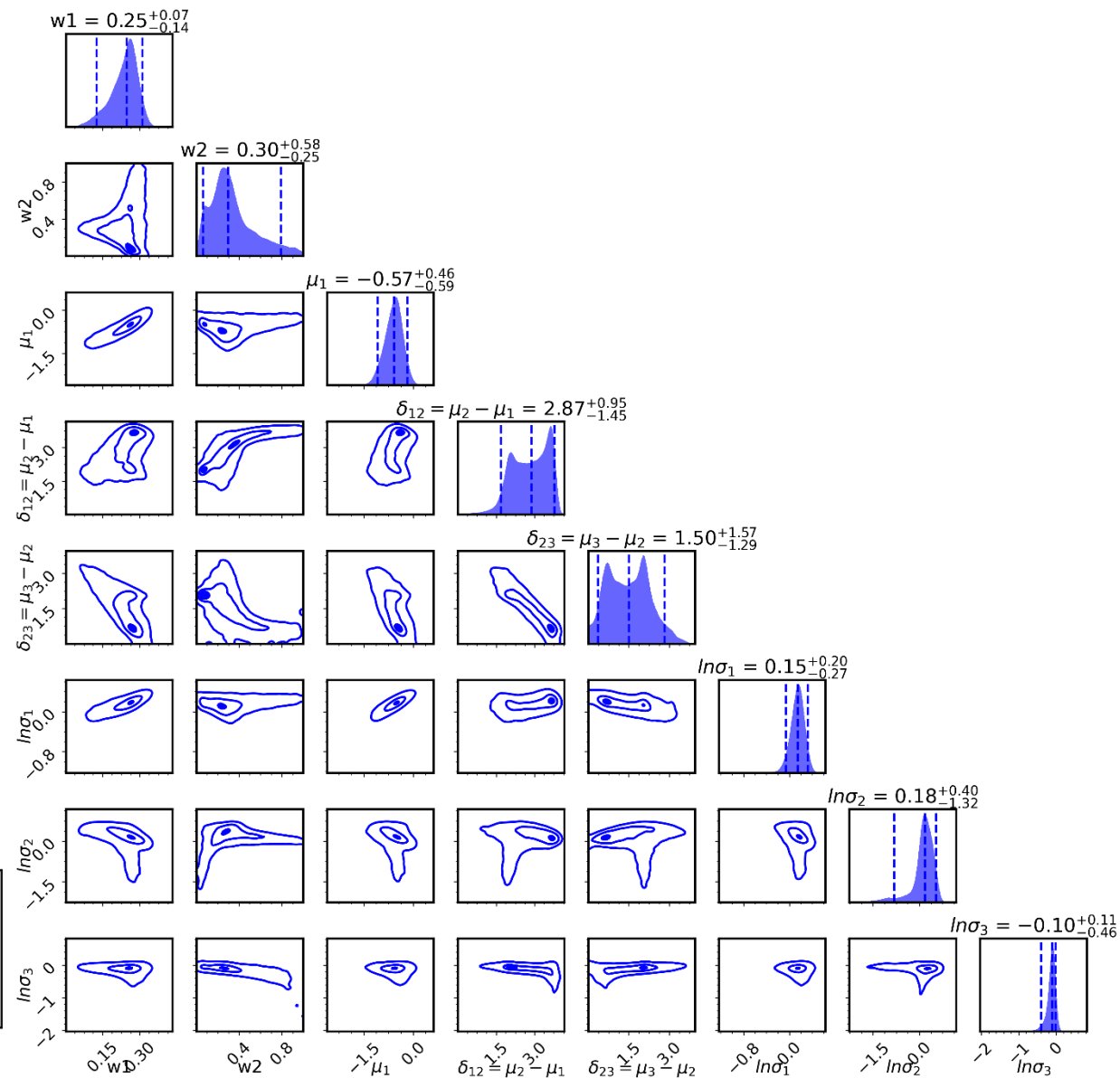
- The evidences  $Z_i$  were computed **5 times** for each hypothesis.
- Validation on **simulated data**:
  - Data generated from **2 components**:  
 $\log\text{BF} = -2.63 \pm 0.19 < 0$   
 $\Rightarrow p(H_0|\mathbf{D}) > p(H_1|\mathbf{D})$
  - Data generated from **3 components**:  
 $\log\text{BF} = 6.07 \pm 0.18 > 0$   
 $\Rightarrow p(H_0|\mathbf{D}) < p(H_1|\mathbf{D})$
- On real data:  
 $\log\text{BF} = -1.75 \pm 0.12 < 0$   
 $\Rightarrow p(H_0|\mathbf{D}) > p(H_1|\mathbf{D})$



Corner plot - Model  $H_0$   
(Metropolis posterior means in red)



Corner plot – Model  $H_1$



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# PROBLEM 5 - CLUSTERING

- **Problem 5:** Another possibility is to classify GRBs in **soft** and **hard**, according to the hardness ratio:

$$HR = \frac{F_{50-100 \text{ keV}}}{F_{20-50 \text{ keV}}}$$

where  $F$  is the measured flux in a certain energy interval. Study the bimodal distribution in the  $\log(T90) - \log(HR)$  space.



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# GIBBS SAMPLING

- A **Markov Chain Monte Carlo (MCMC)** method to draw samples from a **joint posterior** by iteratively sampling each variable from its **conditional distribution** given the current values of the others.
- **Assignments:**  $\mathbf{z} = (z_1, \dots, z_N)$ ,  $z_i \in \{1, \dots, K\}$      $K = \text{\#Gaussian components}$
- **Parameters:** weights  $\mathbf{w} = (w_1, \dots, w_K)$ , gaussian parameters  $\boldsymbol{\theta} = (\boldsymbol{\mu}, \boldsymbol{\sigma}^2) = (\{\mu_k\}_{k=1}^K, \{\sigma_k^2\}_{k=1}^K)$
- Principle:

$$p(\gamma_j | \gamma_{-j}, \mathbf{x}) = p(\gamma_j, \gamma_{-j} | \mathbf{x}) / p(\gamma_{-j} | \mathbf{x}) \propto p(\gamma_j, \gamma_{-j} | \mathbf{x}) = p(\boldsymbol{\gamma} | \mathbf{x})$$

$$\begin{aligned} p(\boldsymbol{\gamma} | \mathbf{x}) &= p(\boldsymbol{\theta}, \mathbf{w}, \mathbf{z} | \mathbf{x}) \propto p(\mathbf{x} | \boldsymbol{\theta}, \mathbf{w}, \mathbf{z}) p(\mathbf{z} | \mathbf{w}) p(\mathbf{w}) p(\boldsymbol{\theta}) = \\ &= \left( \prod_{i=1}^N p(x_i | z_i, \boldsymbol{\theta}) \right) \left( \prod_{i=1}^N w_{z_i} \right) p(\mathbf{w}) p(\boldsymbol{\theta}) = p(\boldsymbol{\theta}) p(\mathbf{w}) \prod_{i=1}^N w_{z_i} \mathcal{N}(x_i | \mu_{z_i}, \sigma_{z_i}^2) \end{aligned}$$

- For each  $\gamma_j$  compute the posterior discarding all the pieces that depends only on  $\gamma_{-j}$ .
-

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# GIBBS SAMPLING ALGORITHM

1. Initialize  $\boldsymbol{\theta}^{(0)}$ ,  $\boldsymbol{w}^{(0)}$  and  $\boldsymbol{z}^{(0)}$ .
  2. For iterations  $t = 1, 2, \dots, M$ :
    - For each  $i$ : draw  $z_i^{(t)} \sim p(z_i | x_i, \boldsymbol{w}^{(t-1)}, \boldsymbol{\theta}^{(t-1)})$ .
    - Draw  $\boldsymbol{w}^{(t)} \sim p(\boldsymbol{w} | \boldsymbol{z}^{(t)})$ .
    - For each  $k$ : draw  $(\mu_k^{(t)}, \sigma_k^{2(t)}) \sim p(\mu_k, \sigma_k^2 | \boldsymbol{x}_k)$  where  $\boldsymbol{x}_k = \{x_i : z_i^{(t)} = k\}$
  3. Remove burn-in and apply thinning.
-

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# CONDITIONAL PROBABILITIES (1/2)

1. For each  $k$ :  $p(z_i = k | x_i, \mathbf{w}, \boldsymbol{\theta}) = \frac{w_k \mathcal{N}(x_i | \mu_k, \sigma_k^2)}{\sum_{h=1}^K w_h \mathcal{N}(x_i | \mu_h, \sigma_h^2)}$ 
    - Draw one component with `numpy.random.choice` using probabilities  $p = \{p(z_i = k | x_i, \mathbf{w}, \boldsymbol{\theta})\}_{k=1}^K$ .
  2.  $p(\mathbf{w} | \mathbf{z}) \propto p(\mathbf{z} | \mathbf{w}) p(\mathbf{w}) = \left( \prod_{i=1}^N w_{z_i} \right) \text{Dir}(\mathbf{w} | \alpha / K) \propto \left( \prod_{k=1}^K w_k^{n_k} w_k^{\alpha / K - 1} \right) \propto \text{Dir}(\mathbf{w} | \alpha / K + n_k)$ 
    - $n_k$  is the number of points assigned to component  $k$ .
    - The prior is a **Dirichlet distribution** with hyperparameter  $\alpha / K$  for all components:  $p(\mathbf{w}) \rightarrow p(\mathbf{w} | \alpha)$ .
  3. Prior:  $p(\theta_k) = p(\mu_k) p(\sigma_k^2) \propto 1 / \sigma_k^2$  where  $p(\mu_k) \propto 1$ ,  $p(\log(\sigma_k^2)) \propto 1 \Rightarrow p(\sigma_k^2) \propto 1 / \sigma_k^2$ 
    - $p(\mu_k, \sigma_k^2 | \mathbf{x}_k) \propto p(\mu_k) p(\sigma_k^2) p(\mathbf{x}_k | \mu_k, \sigma_k^2) =$ 
$$= \frac{1}{\sigma_k^2} \prod_{i=1}^{n_k} \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{1}{2} \frac{(x_{k,i} - \mu_k)^2}{\sigma_k^2}\right) \propto (\sigma_k^2)^{-(\frac{n_k}{2} + 1)} \exp\left(-\frac{1}{2\sigma_k^2} \sum_{i=1}^{n_k} (x_{k,i} - \mu_k)^2\right)$$
-

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# CONDITIONAL PROBABILITIES (2/2)

- Marginalize over  $\mu_k$  to obtain a distribution only in  $\sigma_k^2$ .

$$\begin{aligned} p(\sigma_k^2 | \mathbf{x}_k) &= \int_{-\infty}^{+\infty} p(\mu_k, \sigma_k^2 | \mathbf{x}_k) d\mu_k \propto (\sigma_k^2)^{-\left(\frac{n_k}{2}+1\right)} e^{-\frac{S_k}{2\sigma_k^2}} \int_{-\infty}^{+\infty} \exp\left(-\frac{n_k}{2\sigma_k^2} (\bar{x}_k - \mu_k)^2\right) d\mu_k \propto \\ &\propto (\sigma_k^2)^{-\left(\frac{n_k}{2}+1\right)} e^{-\frac{S_k}{2\sigma_k^2}} (\sigma_k^2)^{\frac{1}{2}} = (\sigma_k^2)^{-\left(\frac{n_k-1}{2}+1\right)} e^{-\frac{S_k}{2\sigma_k^2}} \propto \text{Inv - Gamma}(\sigma_k^2 | \alpha_k = \frac{n_k-1}{2}, \beta_k = \frac{S_k}{2}). \end{aligned}$$

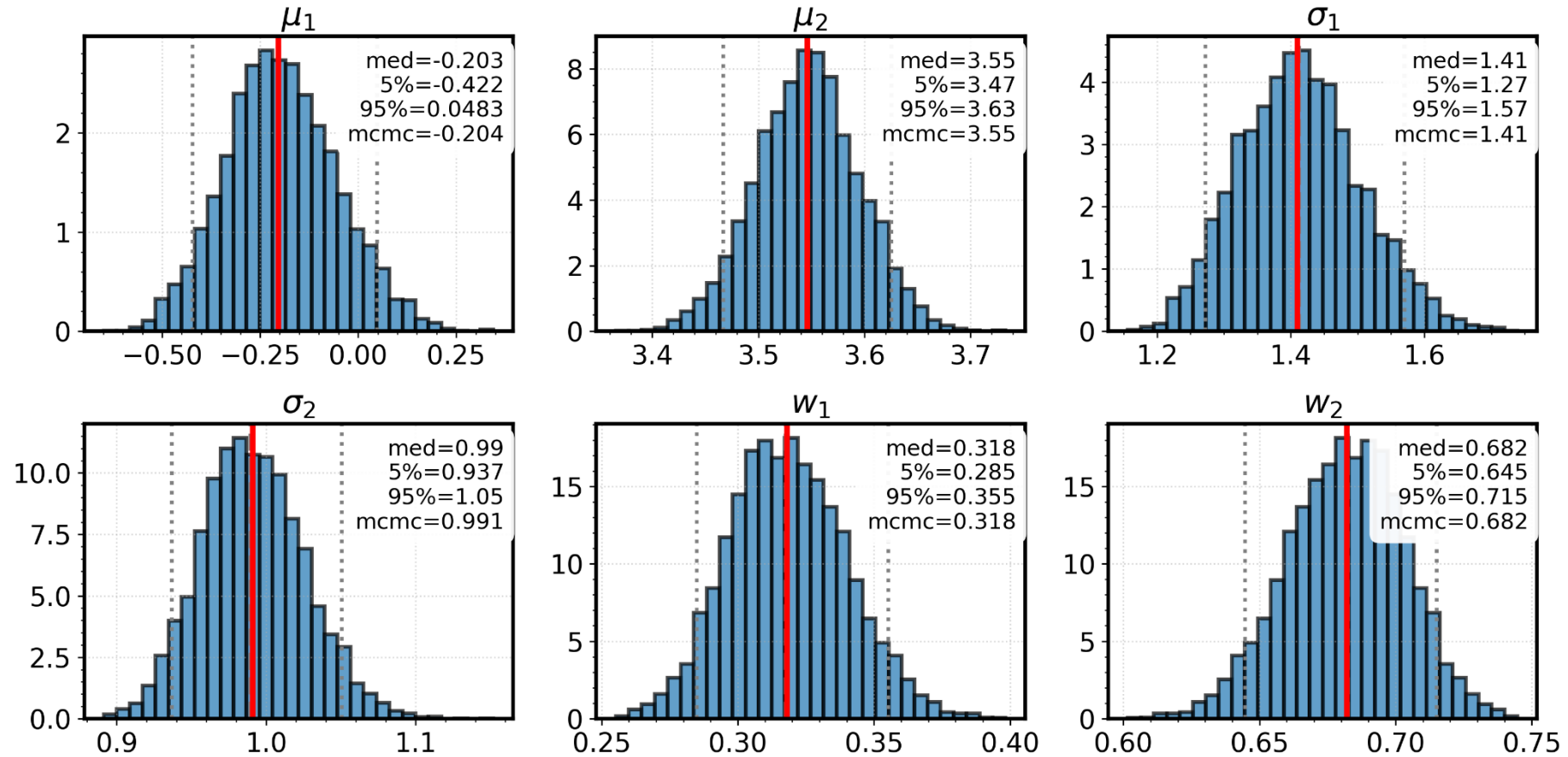
- The relation used is  $\sum_i (x_{k,i} - \mu_k)^2 = S_k + n_k (\bar{x}_k - \mu_k)^2$ , where  $\bar{x}_k = (\sum_i x_{k,i})/n_k$  and  $S_k = \sum_i (x_{k,i} - \bar{x}_k)^2$ .
  - Once  $\sigma_k^2$  is sampled, sample  $\mu_k$  from its conditional distribution:
    - $p(\mu_k | \sigma_k^2, \mathbf{x}_k) \propto \exp\left(-\frac{n_k}{2\sigma_k^2} (\bar{x}_k - \mu_k)^2\right) \propto \mathcal{N}(\mu_k | \bar{x}_k, \sigma_k^2/n_k)$
- $$\Rightarrow p(\mu_k, \sigma_k^2 | \mathbf{x}_k) \propto p(\mu_k | \sigma_k^2, \mathbf{x}_k) p(\sigma_k^2 | \mathbf{x}_k) = \mathcal{N}(\mu_k | \bar{x}_k, \sigma_k^2/n_k) \times \text{Inv - Gamma}\left(\sigma_k^2 | \alpha_k = \frac{n_k-1}{2}, \beta_k = \frac{S_k}{2}\right).$$
- This result requires  $\alpha > 0 \Rightarrow n_k > 1$ , otherwise Inv-Gamma normalization diverges.
-



# RESULTS

(Metropolis posterior means in red)

Gibbs sampling



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# COLLAPSED GIBBS SAMPLING

- If we are interested only in the assignment (clustering) of the data, we can marginalize out all model parameters and compute the conditional distribution of  $z$  given the data.

$$\begin{aligned} \bullet \quad p(z_i = k | z_{\neg i}, \mathbf{x}, \alpha) &= \frac{p(z_i = k, x_i | z_{\neg i}, \mathbf{x}_{\neg i}, \alpha)}{p(x_i | z_{\neg i}, \mathbf{x}_{\neg i}, \alpha)} \propto p(z_i = k, x_i | z_{\neg i}, \mathbf{x}_{\neg i}, \alpha) = \\ &= p(x_i | z_i = k, z_{\neg i}, \mathbf{x}_{\neg i}, \alpha) p(z_i = k | z_{\neg i}, \mathbf{x}_{\neg i}, \alpha) \end{aligned}$$

$$\bullet \quad p(z_i = k | z_{\neg i}, \mathbf{x}_{\neg i}, \alpha) = \int p(z_i = k | \mathbf{w}, \cancel{z_{\neg i}}, \cancel{\mathbf{x}_{\neg i}}, \alpha) p(\mathbf{w} | z_{\neg i}, \cancel{\mathbf{x}_{\neg i}}, \alpha) d\mathbf{w} = p(z_i = k | z_{\neg i}, \alpha)$$

$$\begin{aligned} \bullet \quad p(x_i | z_i = k, z_{\neg i}, \mathbf{x}_{\neg i}, \alpha) &= \iint p(x_i | \mu_k, \sigma_k^2, z_i = k, \cancel{z_{\neg i}}, \cancel{\mathbf{x}_{\neg i}}, \alpha) p(\mu_k, \sigma_k^2 | z_i = k, \cancel{z_{\neg i}}, \cancel{\mathbf{x}_{\neg i}}, \alpha) d\mu_k d\sigma_k^2 = \\ &= p(x_i | z_i = k, \mathbf{x}_{\neg i}) = p(x_i | \{x_j : z_j = k, j \neq i\}) \end{aligned}$$

$$\Rightarrow p(z_i = k | z_{\neg i}, \mathbf{x}, \alpha) \propto p(z_i = k | z_{\neg i}, \alpha) p(x_i | \{x_j : z_j = k, j \neq i\})$$

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# MARGINALIZE OVER THE WEIGHTS

- $p(z_i = k | z_{\neg i}, \alpha) \rightarrow$  Probability that point  $i$  belongs to cluster  $k$  given the assignments of all other points.
- The Dirichlet normalizing constant is:  $B(\alpha) = \int \prod_{j=1}^K w_j^{\alpha_j - 1} d\mathbf{w} = \frac{\prod_{j=1}^K \Gamma(\alpha_j)}{\Gamma(\sum_{j=1}^K \alpha_j)}$
- $p(\mathbf{z} | \alpha) = \int p(\mathbf{z} | \mathbf{w}) p(\mathbf{w} | \alpha) d\mathbf{w} = \frac{1}{B(\alpha/K)} \int \prod_{j=1}^K w_j^{\frac{\alpha}{K} + n_j - 1} d\mathbf{w} = \frac{B(\alpha/K + \mathbf{n})}{B(\alpha/K)}$ , where  $\mathbf{n} = (n_1, \dots, n_K)$
- $p(z_i = k | z_{\neg i}, \alpha) = \frac{p(\mathbf{z} | \alpha)}{p(z_{\neg i} | \alpha)} = \frac{B(\alpha/K + \mathbf{n})}{B(\alpha/K + \mathbf{n}_{\neg i})}$ 
  - where  $n_{j \neg i}$  is the number of points in component  $j$  excluding point  $i$  (if point  $i$  is not in component  $j$ :  $n_{j \neg i} = n_j$ )

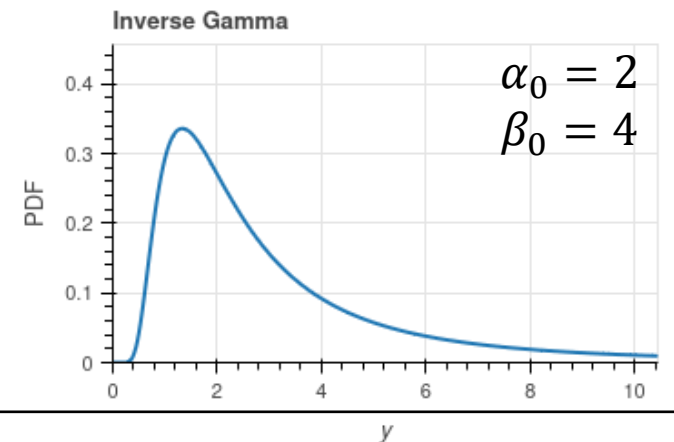
$$\Rightarrow p(z_i = k | z_{\neg i}, \alpha) = \frac{B(\alpha/K + \mathbf{n})}{B(\alpha/K + \mathbf{n}_{\neg i})} = \frac{\Gamma(\alpha/K + n_k)}{\Gamma(\alpha/K + n_{k \neg i})} = \frac{n_{k \neg i} + \alpha/K}{N - 1 + \alpha}$$

- The probability that  $z_i = k$  is proportional to the number of points in components  $k$  ( $n_{k \neg i}$ ).
-

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# MARGINALIZE OVER $\mu_k$ AND $\sigma_k^2$ (1/3)

- $p(x_i | \{x_j: z_j = k, j \neq i\}) = p(x_i | \mathbf{x}_{k \neg i})$  where  $\mathbf{x}_{k \neg i} \stackrel{\text{def}}{=} \{x_j: z_j = k, j \neq i\}$ .
- $p(x_i | \mathbf{x}_{k \neg i}) = \iint p(x_i | \mu_k, \sigma_k^2) p(\mu_k, \sigma_k^2 | \mathbf{x}_{k \neg i}) d\mu_k d\sigma_k^2$ 
  1.  $p(\mu_k, \sigma_k^2 | \mathbf{x}_{k \neg i}) \propto p(\mathbf{x}_{k \neg i} | \mu_k, \sigma_k^2) p(\mu_k, \sigma_k^2)$  where  $p(\mathbf{x}_{k \neg i} | \mu_k, \sigma_k^2) = \prod_{j=1}^{n_{k \neg i}} \mathcal{N}(x_{k \neg i, j} | \mu_k, \sigma_k^2)$
- If we want to create new clusters ( $n_k = 0$ ), we must change the prior on  $\mu_k$  and  $\sigma_k^2$ .
- $p(\mu_k, \sigma_k^2) = p(\mu_k | \sigma_k^2) p(\sigma_k^2) = \mathcal{N}(\mu_k | \mu_0, \sigma_k^2 / \kappa_0) \cdot \text{Inv-Gamma}(\sigma_k^2 | \alpha_0, \beta_0) \stackrel{\text{def}}{=} \text{NIG}(\mu_k, \sigma_k^2 | \mu_0, \kappa_0, \alpha_0, \beta_0)$ 
  - $\mu_0 = (\max(x) + \min(x))/2 \rightarrow$  middle range.
  - $\kappa_0 = 0.01 \rightarrow$  small value to increase the prior variance (uninformative prior).
  - $\alpha_0 = 2, \beta_0 = 4 \rightarrow$  to have good probability in  $\sigma_k^2 \in [1, 4]$ .



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## MARGINALIZE OVER $\mu_k$ AND $\sigma_k^2$ (2/3)

- $\text{NIG}(\mu_k, \sigma_k^2 | \mu_0, \kappa_0, \alpha_0, \beta_0)$  is a **conjugate prior** of  $p(x_{k-i} | \mu_k, \sigma_k^2)$ .
- $\Rightarrow p(\mu_k, \sigma_k^2 | \mathbf{x}_{k-i}) \propto \left( \prod_{j=1}^{n_{k-i}} \mathcal{N}(x_{k-i,j} | \mu_k, \sigma_k^2) \right) \cdot \text{NIG}(\mu_k, \sigma_k^2 | \mu_0, \kappa_0, \alpha_0, \beta_0) = \text{NIG}(\mu_k, \sigma_k^2 | \mu_n, \kappa_n, \alpha_n, \beta_n)$

$$\bullet \begin{cases} \alpha_n = \alpha_0 + \frac{n_{k-i}}{2}, & \beta_n = \beta_0 + \frac{S_{k-i}}{2} + \frac{\kappa_0 n_{k-i}}{2\kappa_n} (\bar{x}_{k-i} - \mu_0)^2 \\ \kappa_n = \kappa_0 + n_{k-i}, & \mu_n = \frac{\kappa_0 \mu_0 + n_{k-i} \bar{x}_{k-i}}{\kappa_0 + n_{k-i}} \end{cases}$$

$$2. \quad p(x_i | \mu_k, \sigma_k^2) = \mathcal{N}(x_i | \mu_k, \sigma_k^2), \quad p(\mu_k, \sigma_k^2 | \mathbf{x}_{k-i}) = \mathcal{N}(\mu_k | \mu_n, \sigma_k^2 / \kappa_n) \cdot \text{Inv-Gamma}(\sigma_k^2 | \alpha_n, \beta_n)$$

- Let's put the pieces together, isolate the components dependent on  $\mu_k$ , and apply the convolution of two Gaussians (solution problem 1b: sum of variances).

$$\bullet \Rightarrow p(x_i | \mathbf{x}_{k-i}) = \int \left( \int \mathcal{N}(x_i | \mu_k, \sigma_k^2) \cdot \mathcal{N}(\mu_k | \mu_n, \sigma_k^2 / \kappa_n) d\mu_k \right) \text{Inv-Gamma}(\sigma_k^2 | \alpha_n, \beta_n) d\sigma_k^2 = \\ = \int \mathcal{N}(x_i | \mu_n, \sigma_k^2 (1 + 1/\kappa_n)) \cdot \text{Inv-Gamma}(\sigma_k^2 | \alpha_n, \beta_n) d\sigma_k^2$$

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## MARGINALIZE OVER $\mu_k$ AND $\sigma_k^2$ (3/3)

3. The Inverse-Gamma normalizing constant is:  $\int x^{-(\alpha+1)} e^{-\beta/x} dx = \Gamma(\alpha)\beta^{-\alpha}$ .

- Let's write explicitly the distribution inside the integral and take out everything that does not depend on  $\sigma_k^2$ .

$$\begin{aligned} p(x_i | \mathbf{x}_{k \neg i}) &= \frac{\beta_n^{\alpha_n}}{\Gamma(\alpha_n)} \frac{1}{\sqrt{2\pi(1+1/\kappa_n)}} \int (\sigma_k^2)^{-(\alpha_n+1/2+1)} \exp\left(-\frac{1}{\sigma_k^2} \left(\beta_n + \frac{(x_i - \mu_n)^2}{2(1+1/\kappa_n)}\right)\right) d\sigma_k^2 = \\ &= \frac{\beta_n^{\alpha_n}}{\Gamma(\alpha_n)} \frac{1}{\sqrt{2\pi(1+1/\kappa_n)}} \Gamma(\alpha_n + 1/2) \left(\beta_n + \frac{(x_i - \mu_n)^2}{2(1+1/\kappa_n)}\right)^{-(\alpha_n+1/2)} \end{aligned}$$

- Let's define  $\nu \stackrel{\text{def}}{=} 2\alpha_n$  and  $s^2 \stackrel{\text{def}}{=} \frac{\beta_n}{\alpha_n} \left(1 + \frac{1}{\kappa_n}\right)$ :

$$\Rightarrow p(x_i | \mathbf{x}_{k \neg i}) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right) \sqrt{\pi\nu s^2}} \left(1 + \frac{(x_i - \mu_n)^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}} \stackrel{\text{def}}{=} t_\nu(x_i | \mu_n, s) \quad \textbf{t-Student distribution}$$

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# COLLAPSED GIBBS SAMPLING ALGORITHM

1. Initialize  $\boldsymbol{\theta}^{(0)}$ ,  $\boldsymbol{w}^{(0)}$  and  $\boldsymbol{z}^{(0)}$ , and choose the hyperparameters  $\alpha, \mu_0, \kappa_0, \alpha_0$  and  $\beta_0$ .
2. For iterations  $t = 1, 2, \dots, M$ :
  - For each  $i$ :
    - Remove point  $i$  from the cluster it occupied in the previous iteration  $t - 1$ .
    - Draw  $z_i^{(t)} \sim p(z_i = k | \boldsymbol{z}_{-i}, \boldsymbol{x}, \alpha) \propto (n_{k \neg i} + \alpha/K) \cdot t_\nu(x_i | \mu_n, s)$ .

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# NON-PARAMETRIC COLLAPSED GIBBS SAMPLING

- Let us assume the number of clusters  $K$  is unknown, so we adopt a nonparametric model that permits an **unbounded number of clusters** ( $K \rightarrow \infty$ ).
  - The only part that depends on  $K$  is  $p(z_i = k|z_{\neg i}, \alpha)$ .
  - $p(z_i = k|z_{\neg i}, \alpha) = \lim_{K \rightarrow \infty} \frac{n_{k \neg i} + \alpha/K}{N-1+\alpha}$ 
    - $n_{k \neg i} > 0 \Rightarrow p(z_i = k|z_{\neg i}, \alpha) = \frac{n_{k \neg i}}{N-1+\alpha}$
    - $n_{k \neg i} = 0 \Rightarrow \begin{aligned} p(z_i = \text{new}|z_{\neg i}, \alpha) &= 1 - p(z_i = \text{old}|z_{\neg i}, \alpha) = 1 - \sum_{k=1}^{K^*} p(z_i = k|z_{\neg i}, \alpha) = \\ &= 1 - \sum_{k=1}^{K^*} \frac{n_{k \neg i}}{N-1+\alpha} = 1 - \frac{N-1}{N-1+\alpha} = \frac{\alpha}{N-1+\alpha} \end{aligned}$ 
      - $K^*$  denotes the number of occupied clusters, each containing at least one observation.
  - $p(z_i = k|z_{\neg i}, \alpha) = \frac{1}{N-1+\alpha} \cdot \begin{cases} n_{k \neg i} & \text{if } n_{k \neg i} > 0 \\ \alpha & \text{if } n_{k \neg i} = 0 \end{cases} \Rightarrow \text{Chinese restaurant process}$
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# NONPARAMETRIC ALGORITHM

1. Initialize  $\boldsymbol{\theta}^{(0)}$ ,  $\boldsymbol{w}^{(0)}$  and  $\boldsymbol{z}^{(0)}$  and choose the hyperparameters:  $\alpha, \mu_0, \kappa_0, \alpha_0$  and  $\beta_0$ .
  2. For iterations  $t = 1, 2, \dots, M$ :
    - For each  $i$ :
      - Remove point  $i$  from the cluster it occupied in the previous iteration  $t - 1$ .
      - If the cluster becomes empty, remove it.
      - For each  $k = 1, \dots, K^*$ : compute  $p(z_i = k | z_{-i}, \boldsymbol{x}, \alpha) \propto n_{k \neg i} \cdot t_v(x_i | \mu_n, s) \stackrel{\text{def}}{=} q_k$ .
      - Compute  $p(z_i = \text{new} | z_{-i}, \boldsymbol{x}, \alpha) \propto \alpha \cdot t_v(x_i | \mu_n, s) \stackrel{\text{def}}{=} q_{\text{new}}$ .
      - Normalize:  $p_k = q_k / (q_{\text{new}} + \sum_{k=1}^{K^*} q_k)$  and  $p_{\text{new}} = q_{\text{new}} / (q_{\text{new}} + \sum_{k=1}^{K^*} q_k)$
      - Draw  $z_i^{(t)}$  with `numpy.random.choice` using probabilities  $p = (p_1, \dots, p_{K^*}, p_{\text{new}})$
      - If  $z_i^{(t)} = \text{new}$ , create a new cluster and assign  $i$  to it;  
otherwise assign  $i$  to the selected existing cluster.
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# MULTIDIMENSIONAL GENERALIZATION (1/3)

- The data-dependent term is  $p(x_i|\mathbf{x}_{k\rightarrow i})$ , where  $x_i = (x_{i_1}, \dots, x_{i_d})$ ,  $x_i \in \mathbb{R}^d$ .

- $p(x_i|\mathbf{x}_{k\rightarrow i}) = \iint p(x_i|\mu_k, \Sigma_k) p(\mu_k, \Sigma_k|\mathbf{x}_{k\rightarrow i}) d\mu_k d\Sigma_k$

- Here  $\mu_k = (\mu_{k_1}, \dots, \mu_{k_d})$  and  $\Sigma_k$  is the  $d \times d$  covariance matrix.

1.  $p(\mu_k, \Sigma_k|\mathbf{x}_{k\rightarrow i}) \propto p(\mathbf{x}_{k\rightarrow i}|\mu_k, \Sigma_k) p(\mu_k, \Sigma_k)$

- $p(\mathbf{x}_{k\rightarrow i}|\mu_k, \Sigma_k) = \prod_{j=1}^{n_{k\rightarrow i}} \mathcal{N}(x_{k\rightarrow i,j}|\mu_k, \Sigma_k)$ , the likelihood is a products of **multivariate Gaussian**.
  - $p(\mu_k, \Sigma_k) = p(\mu_k|\Sigma_k) p(\Sigma_k) = \mathcal{N}(\mu_k|\mu_0, \Sigma_k/\kappa_0) \cdot \text{Inv-Wishart}(\Sigma_k|\nu_0, \Lambda_0) \stackrel{\text{def}}{=} \text{NIW}(\mu_k, \Sigma_k|\mu_0, \kappa_0, \nu_0, \Lambda_0)$ 
    - **Inv-Wishart**( $\Sigma_k|\nu_0, \Lambda_0$ ) is the multivariate analogue of the Inv-Gamma ( $\sigma_k^2|\alpha_0, \beta_0$ ).
    - If  $d = 1 \Rightarrow \nu_0 = 2\alpha_0$ , and  $\Lambda_0 = 2\beta_0$ . For consistency, we choose  $\nu_0 = 4$ , and  $\Lambda_0 = 8 \cdot I_{d \times d}$ .
    - $\kappa_0 = 0.01 \rightarrow$  small value to increase the prior covariance (uninformative prior).
    - $\mu_0 = \left\{ \left( \max(x_{i_h}) + \min(x_{i_h}) \right) / 2 \right\}_{h=1}^d \rightarrow$  middle range.
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# MULTIDIMENSIONAL GENERALIZATION (2/3)

- $\text{NIW}(\mu_k, \Sigma_k | \mu_0, \kappa_0, \nu_0, \Lambda_0)$  is a **conjugate prior** of  $p(x_{k-i} | \mu_k, \Sigma_k)$ .
- $\Rightarrow p(\mu_k, \Sigma_k | \mathbf{x}_{k-i}) \propto \left( \prod_{j=1}^{n_{k-i}} \mathcal{N}(x_{k-i,j} | \mu_k, \Sigma_k) \right) \cdot \text{NIG}(\mu_k, \Sigma_k | \mu_0, \kappa_0, \nu_0, \Lambda_0) = \text{NIG}(\mu_k, \Sigma_k | \mu_n, \kappa_n, \nu_n, \Lambda_n)$ 
  - $$\begin{cases} \nu_n = \nu_0 + n_{k-i}, & \Lambda_n = \Lambda_0 + S_{k-i} + \frac{\kappa_0 n_{k-i}}{\kappa_n} (\bar{\mathbf{x}}_{k-i} - \mu_0)(\bar{\mathbf{x}}_{k-i} - \mu_0)^\top \\ \kappa_n = \kappa_0 + n_{k-i}, & \mu_n = \frac{\kappa_0 \mu_0 + n_{k-i} \bar{\mathbf{x}}_{k-i}}{\kappa_0 + n_{k-i}} \end{cases}$$

2.  $p(x_i | \mu_k, \Sigma_k) = \mathcal{N}(x_i | \mu_k, \Sigma_k), \quad p(\mu_k, \Sigma_k | \mathbf{x}_{k-i}) = \mathcal{N}(\mu_k | \mu_n, \Sigma_k / \kappa_n) \cdot \text{Inv-Wishart}(\Sigma_k | \nu_n, \Lambda_n)$

- Let's put the pieces together, isolate the components dependent on  $\mu_k$ , and apply the convolution of two Gaussians.
  - $$\begin{aligned} \Rightarrow p(x_i | \mathbf{x}_{k-i}) &= \int \left( \int \mathcal{N}(x_i | \mu_k, \Sigma_k) \cdot \mathcal{N}(\mu_k | \mu_n, \Sigma_k / \kappa_n) d\mu_k \right) \text{Inv-Wishart}(\Sigma_k | \nu_n, \Lambda_n) d\Sigma_k = \\ &= \int \mathcal{N}(x_i | \mu_n, \Sigma_k (1 + 1/\kappa_n)) \cdot \text{Inv-Wishart}(\Sigma_k | \nu_n, \Lambda_n) d\Sigma_k \end{aligned}$$
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# MULTIDIMENSIONAL GENERALIZATION (3/3)

3. The Inverse-Wishart normalizing constant is:  $\int |\Sigma|^{-(v+d+1)/2} e^{-\text{tr}(\Lambda \Sigma^{-1})/2} d\Sigma = \left(2^{vd/2} \Gamma_d(v/2)\right) / |\Lambda|^{v/2}$ .

- Let's write explicitly the distribution inside and define  $c \stackrel{\text{def}}{=} \kappa_n / (\kappa_n + 1)$  and  $u \stackrel{\text{def}}{=} x_i - \mu_n$ .

$$p(x_i | \mathbf{x}_{k \neg i}) = (2\pi)^{-\frac{d}{2}} \frac{|\Lambda_n|^{v_n/2}}{2^{v_n d/2} \Gamma_d(v_n/2)} \int |\Sigma_k|^{-\frac{1}{2}} |\Sigma_k|^{-\frac{v_n+d+1}{2}} e^{-\frac{1}{2} u^\top \Sigma_k^{-1} u} e^{-\frac{1}{2} \text{tr}(\Lambda_n \Sigma_k^{-1})} d\Sigma_k =$$

- $$\begin{aligned} &= (2\pi)^{-\frac{d}{2}} \frac{|\Lambda_n|^{v_n/2}}{2^{v_n d/2} \Gamma_d(v_n/2)} \int |\Sigma_k|^{-\frac{(v_n+1)+d+1}{2}} e^{-\frac{1}{2} \text{tr}((cuu^\top + \Lambda_n) \Sigma_k^{-1})} d\Sigma_k \\ &= \pi^{-\frac{d}{2}} \frac{\Gamma_d((v_n+1)/2)}{\Gamma_d(v_n/2)} |\Lambda_n|^{-1/2} (1 + cu^\top \Lambda_n^{-1} u)^{-(v_n+d)/2} \end{aligned}$$

- Let's define  $v_t \stackrel{\text{def}}{=} v_n - d + 1$  and  $\Sigma_t \stackrel{\text{def}}{=} \frac{\kappa_n + 1}{\kappa_n v_t} \Lambda_n$ :

$$\Rightarrow p(x_i | \mathbf{x}_{k \neg i}) = \frac{\Gamma\left(\frac{v_t + 1}{2}\right)}{\Gamma\left(\frac{v_t}{2}\right) (\pi v_t)^{d/2} |\Sigma_t|^{1/2}} \left(1 + \frac{1}{v_t} (x_i - \mu_n)^\top \Sigma_t^{-1} (x_i - \mu_n)\right)^{-\frac{v_t+d}{2}} \stackrel{\text{def}}{=} t_{v_t}(x_i | \mu_n, \Sigma_t)$$

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**Multivariate t-distribution**

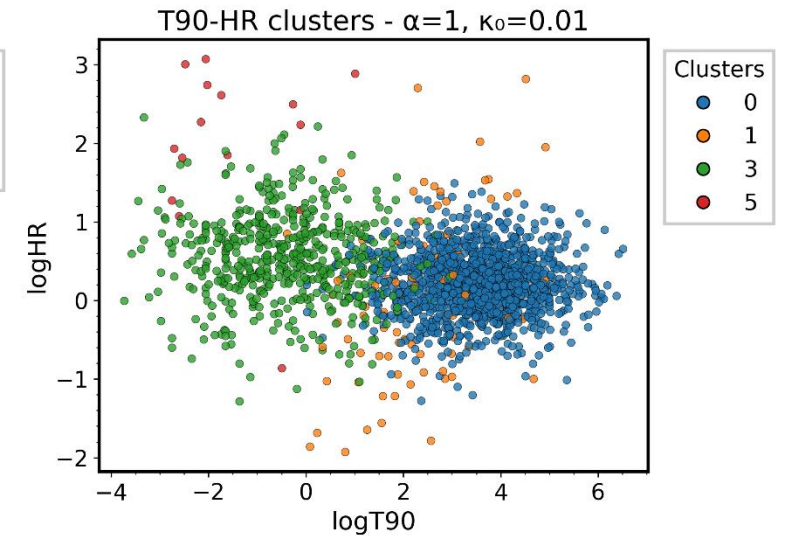
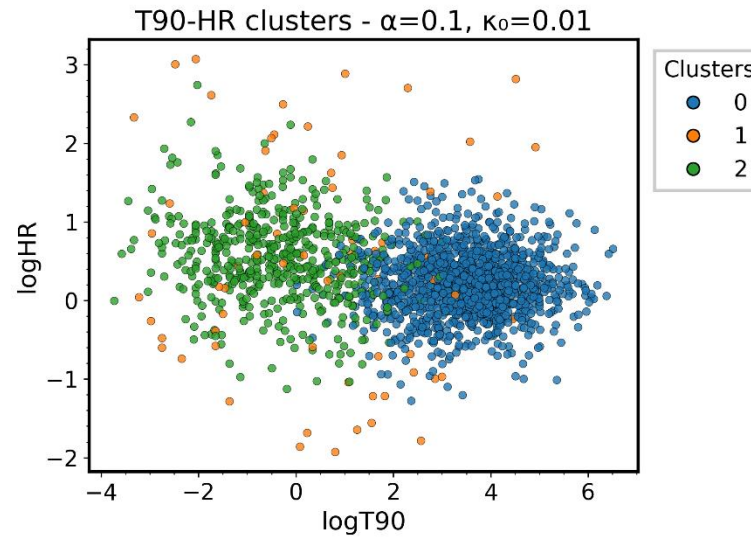
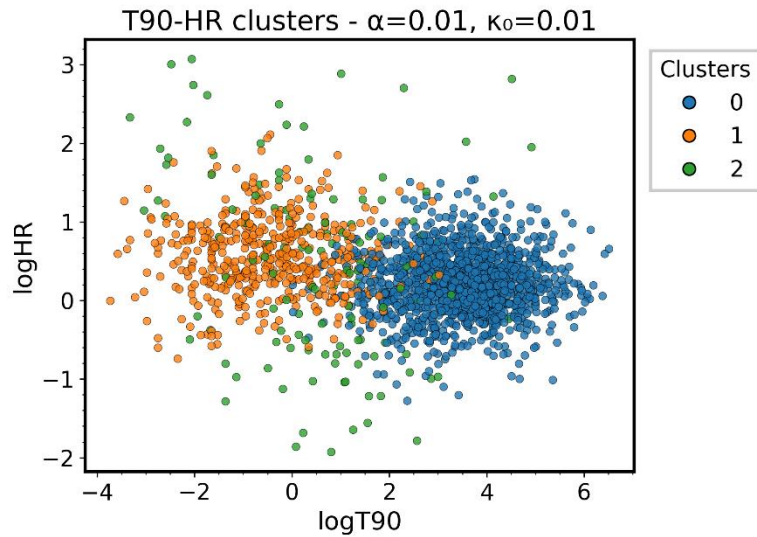
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# MULTIDIMENSIONAL ALGORITHM

1. Initialize  $\boldsymbol{\theta}^{(0)}$ ,  $\boldsymbol{w}^{(0)}$  and  $\boldsymbol{z}^{(0)}$  and choose the hyperparameters:  $\alpha, \mu_0, \kappa_0, \nu_0$  and  $\Sigma_0$ .
  2. For iterations  $t = 1, 2, \dots, M$ :
    - For each  $i$ :
      - Remove point  $i$  from the cluster it occupied in the previous iteration  $t - 1$ .
      - If the cluster becomes empty, remove it.
      - For each  $k = 1, \dots, K^*$ : compute  $p(z_i = k | \boldsymbol{z}_{-i}, \boldsymbol{x}, \alpha) \propto n_{k-i} \cdot t_{\nu_t}(x_i | \mu_n, \Sigma_t) \stackrel{\text{def}}{=} q_k$ .
      - Compute  $p(z_i = \text{new} | \boldsymbol{z}_{-i}, \boldsymbol{x}, \alpha) \propto \alpha \cdot t_{\nu}(x_i | \mu_n, s) \stackrel{\text{def}}{=} q_{\text{new}}$ .
      - Normalize:  $p_k = q_k / (q_{\text{new}} + \sum_{k=1}^{K^*} q_k)$  and  $p_{\text{new}} = q_{\text{new}} / (q_{\text{new}} + \sum_{k=1}^{K^*} q_k)$
      - Draw  $z_i^{(t)}$  with `numpy.random.choice` using probabilities  $p = (p_1, \dots, p_{K^*}, p_{\text{new}})$
      - If  $z_i^{(t)} = \text{new}$ , create a new cluster and assign  $i$  to it; otherwise assign  $i$  to the selected existing cluster.
-

# VARIABLE $\alpha$ RESULTS.

#Iterations = 1000, #Initial clusters=2



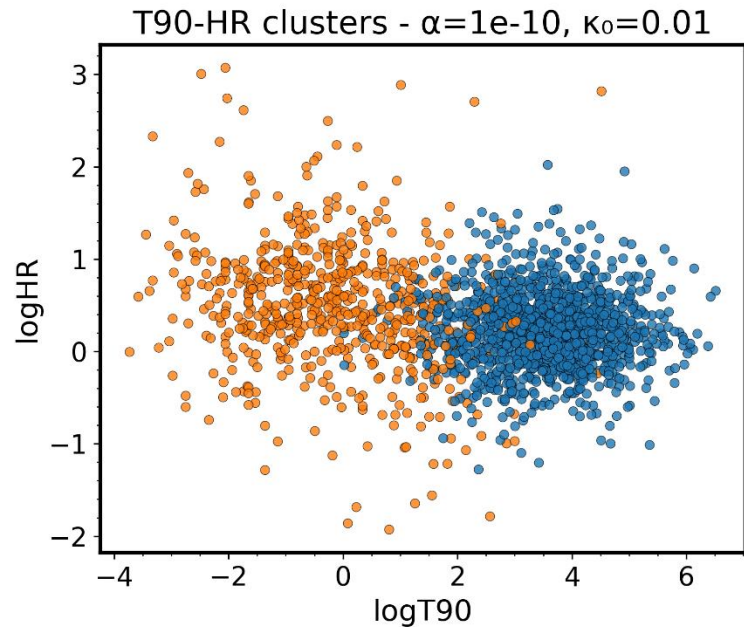
- $\mu_0 = [3.51, 0.212]$ ,  $\sigma_0 = [1.02, 0.395]$
- $\mu_1 = [-0.467, 0.566]$ ,  $\sigma_1 = [1.27, 0.482]$
- $\mu_2 = [-0.541, 0.465]$ ,  $\sigma_2 = [1.68, 1.16]$

- $\mu_0 = [3.50, 0.216]$ ,  $\sigma_0 = [1.02, 0.399]$
- $\mu_1 = [0.458, 0.480]$ ,  $\sigma_1 = [2.06, 1.34]$
- $\mu_2 = [-0.373, 0.547]$ ,  $\sigma_2 = [1.27, 0.554]$

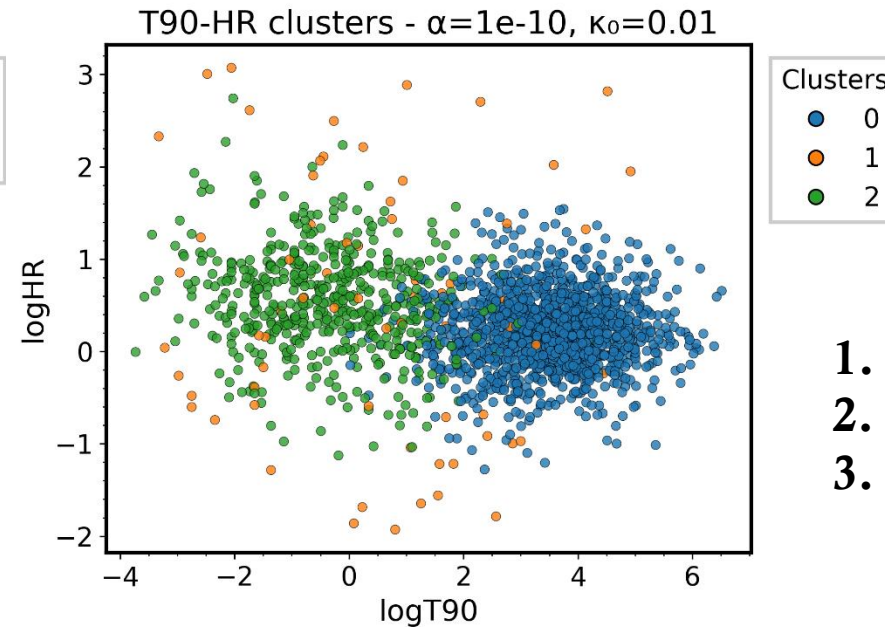
- $\mu_0 = [3.53, 0.216]$ ,  $\sigma_0 = [1.01, 0.38]$
- $\mu_1 = [2.33, 0.075]$ ,  $\sigma_1 = [1.19, 1.00]$
- $\mu_3 = [-0.395, 0.544]$ ,  $\sigma_3 = [1.28, 0.551]$
- $\mu_5 = [-1.51, 1.20]$ ,  $\sigma_5 = [1.16, 0.986]$

# FIXED COMPONENTS.

#Iterations = 1000



- #Initial clusters = 2
- $\mu_0 = [3.55, 0.228]$ ,  $\sigma_0 = [0.987, 0.402]$
- $\mu_1 = [-0.174, 0.495]$ ,  $\sigma_1 = [1.43, 0.692]$



- #Initial clusters = 3
- $\mu_0 = [3.50, 0.216]$ ,  $\sigma_0 = [1.02, 0.399]$
- $\mu_1 = [0.458, 0.480]$ ,  $\sigma_1 = [2.06, 1.34]$
- $\mu_2 = [-0.373, 0.547]$ ,  $\sigma_2 = [1.27, 0.554]$

1. Short GRB  $\rightarrow$  Hard GRB
2. Long GRB  $\rightarrow$  Soft GRB
3. Outliers

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